

第三章 傅里叶光学概要

3.1 F.T的数理基础

3.2 透镜的F.T性质

3.3 透镜成像的傅里叶分析

3.4 光学空间滤波及全光图像处理

一、F.T数理基础

光学傅立叶变换的性质

(一) F.T定理

• 线性定理

若 $F\{f(x, y)\} = F(u, v)$, $F\{g(x, y)\} = G(u, v)$ 。 a, β 为常数

$$F\{af(x, y) + \beta g(x, y)\} = aF\{f(x, y)\} + \beta F\{g(x, y)\}$$

物理意义：

光学系统若同时有几个输入（或激励），其所产生的输出（或响应） **恒等于** 每个输入单独引起的输出之和。

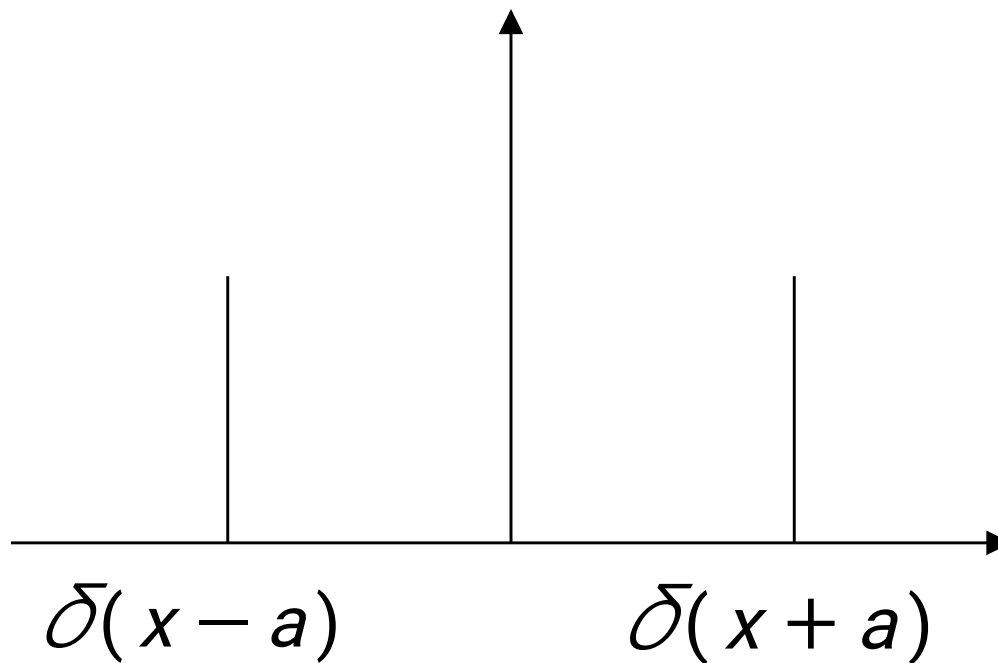
光的线性叠加原理的体现

- 举例：

$$a\delta(x+a) + \beta\delta(x-a)$$



$$ae^{j2\pi ua} + \beta e^{-j2\pi ua}$$



光学傅立叶变换的性质

- 相似定理

$$F \{ U(ax, by) \} = \frac{1}{|ab|} U\left(\frac{f_x}{a}, \frac{f_y}{b}\right)$$

物理意义：物函数的空间坐标伸展（压缩），其频谱函数形式不变，只是频标产生相应的压缩（伸展），并在幅度上有一个整体变化。

如：若夫琅和费衍射的缝宽变窄，则衍射条纹变宽。

光学傅立叶变换的性质

- 相移定理:

$$\mathcal{F} \{ U(x-a, y-b) \} = U(f_x, f_y) e^{-j2\pi(f_x a + f_y b)}$$

物理意义：物函数在空间域中平移，只使频率域中产生一个线性相移

如：夫琅和费衍射条纹的位置并不随源点（或者说衍射屏）的移动而改变。

光学傅立叶变换的性质

- PARSEVA定理:

$$\int \int_{-\infty}^{\infty} |U(x, y)|^2 dx dy = \int \int_{-\infty}^{\infty} |U(f_x, f_y)|^2 df_x df_y$$

波面上的光能

频谱面上的光能

物理意义：当变换透镜表面的反射和玻璃介质的吸收可以忽略时，光能量守恒。

光学傅立叶变换的性质

卷积定理:

$$U(x, y) = U_1(x, y) * U_2(x, y)$$

$$U(x, y) = \int \int_{-\infty}^{\infty} U_1(x', y') U_2(x - x', y - y') dx' dy'$$

↑

反转后平移

$$F\{f(x, y) * g(x, y)\} = F(u, v) \bullet G(u, v)$$

两个函数较繁的卷积运算可以简化为较简单的相乘运算。在描述光学系统的物像关系中，卷积是常用的一种运算。

光学傅立叶变换的性质

- 自相关定理:

$$U(x, y) = U_1(x, y) \otimes U_1(x, y) = \int \int_{-\infty}^{\infty} U_1(x', y') U_1^*(x' - x, y' - y) dx' dy'$$

↑
先平移后共轭

$$F\{f(x, y) \otimes f(x, y)\} = |F(u, v)|^2$$

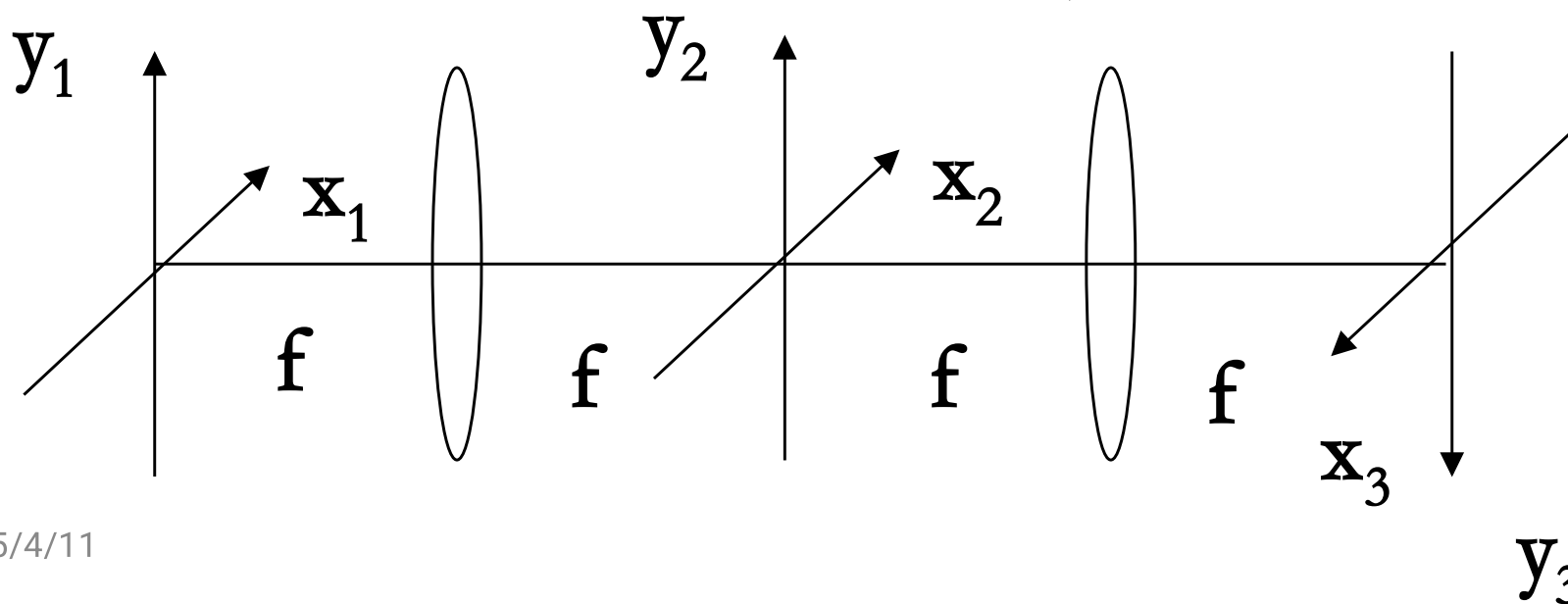
光学傅立叶变换的性质

• (二)、傅里叶积分定理

$$F \{ F \{ U(x, y) \} \} = F \{ U(f_x, f_y) \} = U(-x, -y)$$

$$F \{ F^{-1} \{ U(x, y) \} \} = F \{ U(-f_x, -f_y) \} = U(x, y)$$

物理意义：说明透镜只能做正F.T。



(三) 常用傅里叶变换式

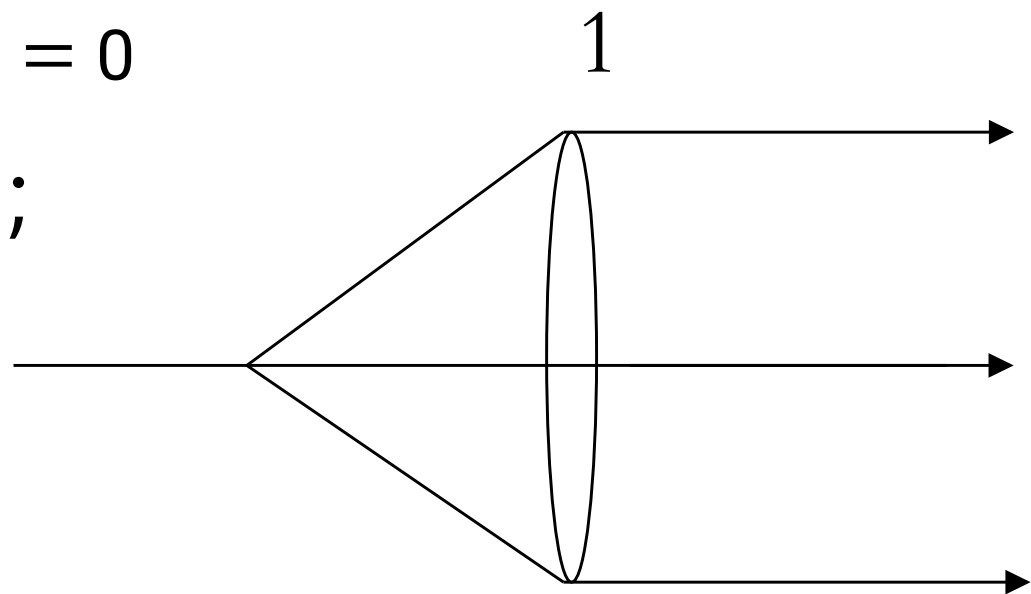
函数

$$\delta(x, y)$$

变换

性质为 $\delta(x, y) = \begin{cases} \infty, & x = y = 0 \\ 0, & \text{其它;} \end{cases}$

$$\iint_{-\infty}^{\infty} \delta(x, y) dx dy = 1.$$



$$F(u, v) = \int_{-\infty}^{+\infty} f(x, y) e^{-j2\pi(ux + vy)} dx dy$$

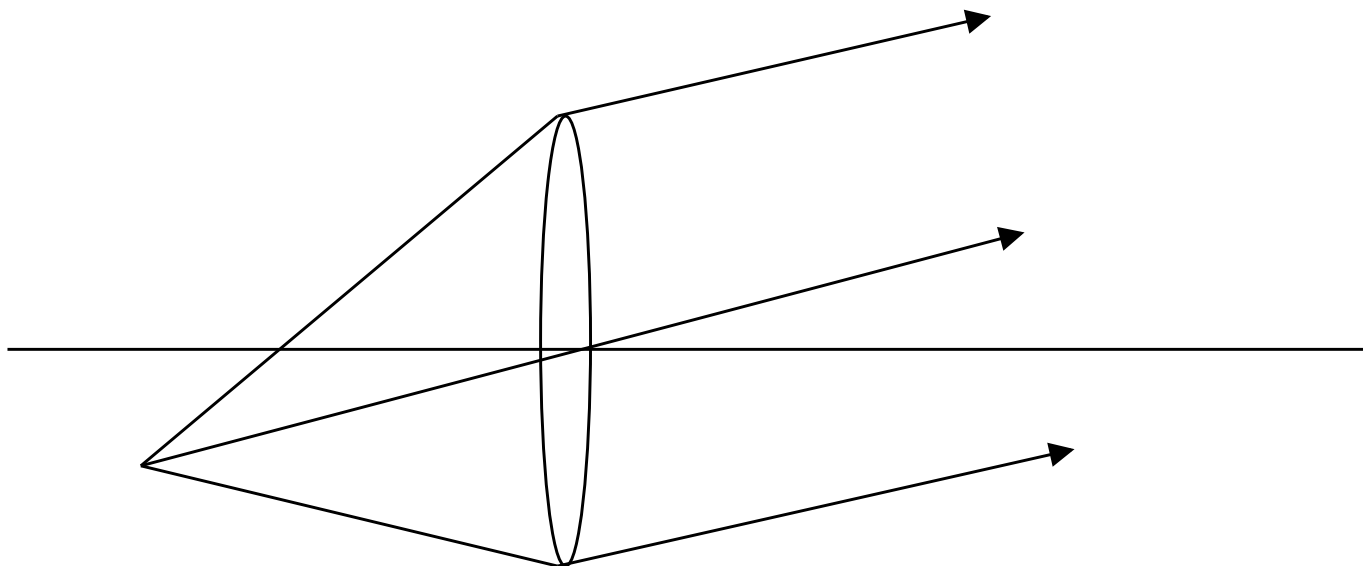
(三)、常用傅里叶变换式

函数

变换

$$\delta\left(f_x - \frac{1}{2}, f_y - \frac{1}{2}\right)$$

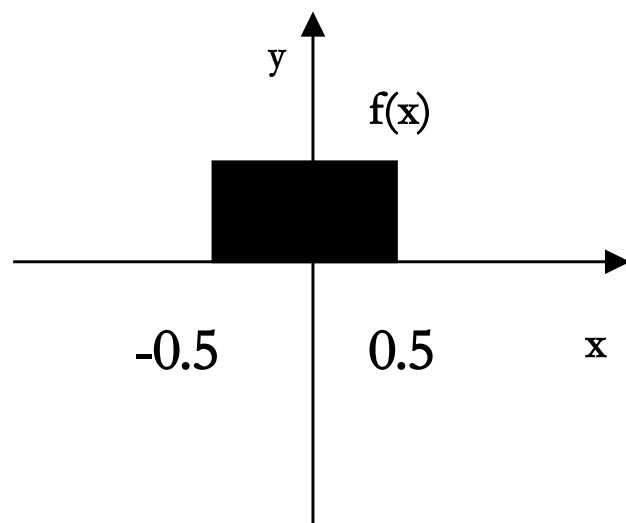
$$e^{-j\pi(x+y)}$$



(三) 常用傅里叶变换式

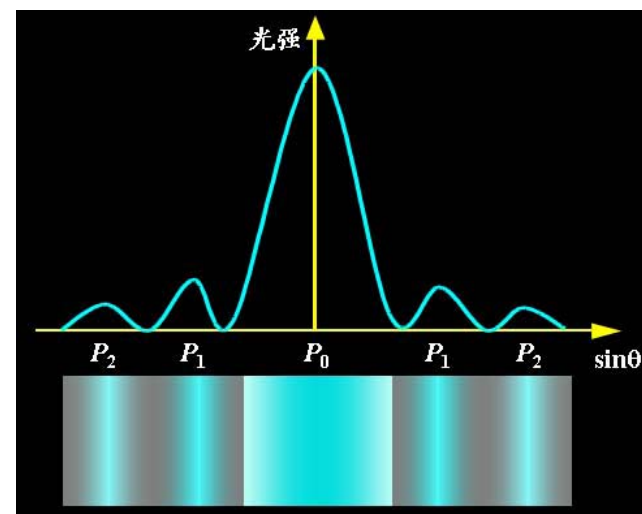
函数

$$\text{rect}(x) = \begin{cases} 1, & 0 \leq |x| \leq \frac{1}{2}; \\ 0, & \text{其它.} \end{cases}$$



变换

$$\sin cu = \frac{\sin \pi u}{\pi u}$$

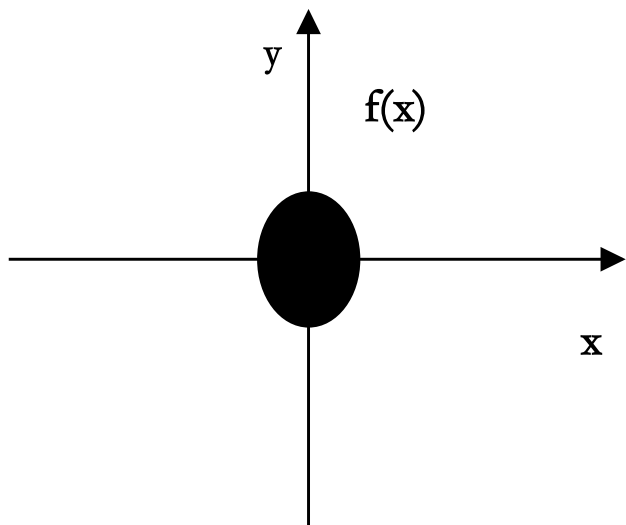


(三) 常用傅里叶变换式

函数

$\text{circ}(r)$

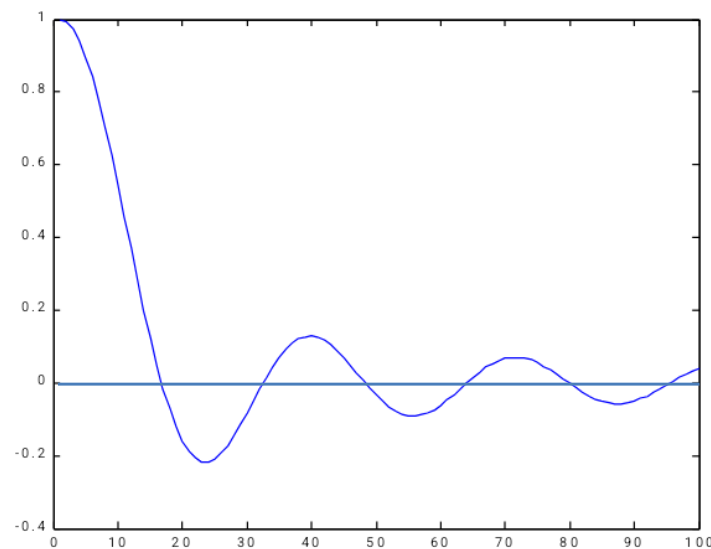
性质为 $\text{circ}(r) = \begin{cases} 1, & 0 \leq r \leq 1, \\ 0, & \text{其它.} \end{cases}$



关系式: $\exp(ix \cos \varphi) = \sum_{n=-\infty}^{\infty} j^n J_n(x) e^{in\varphi},$

变换

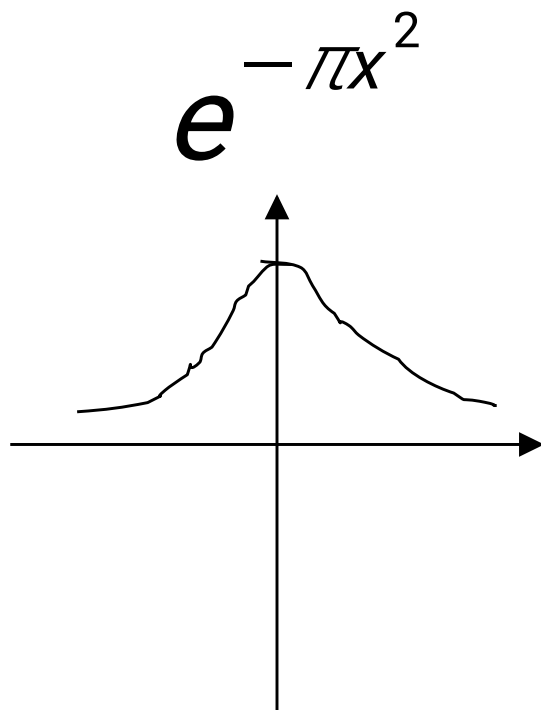
$$\frac{J_1(2\pi\rho)}{\rho}$$



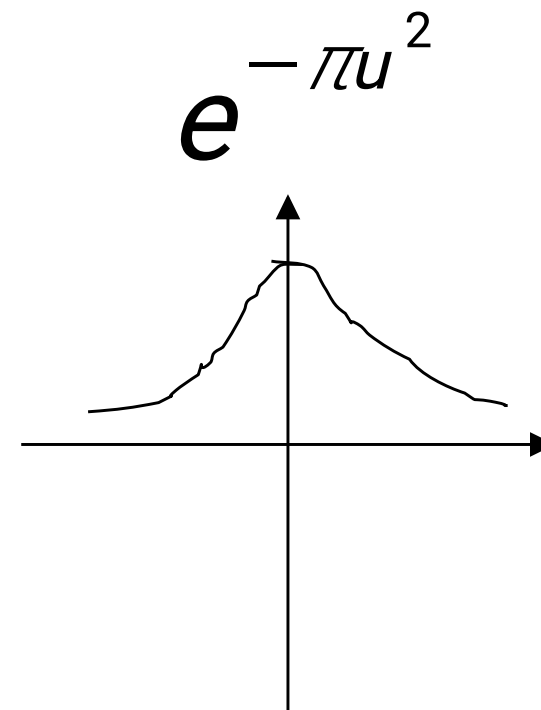
$$\frac{d}{dx}(xJ_1(x)) = xJ_0(x)$$

(三) 常用傅里叶变换式

函数



变换

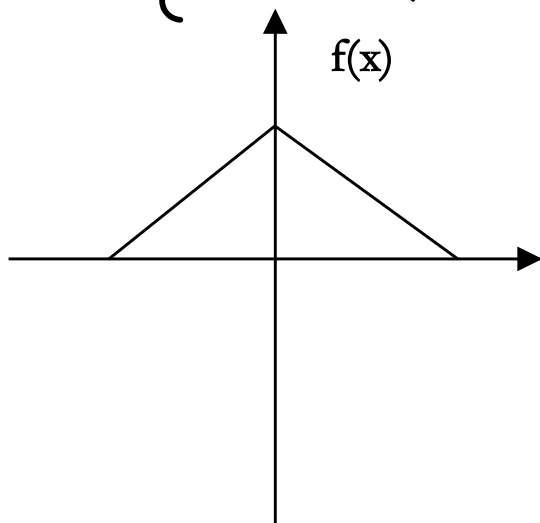


$\therefore$ 激光（高斯型）经过透镜后不变

(三) 常用傅里叶变换式

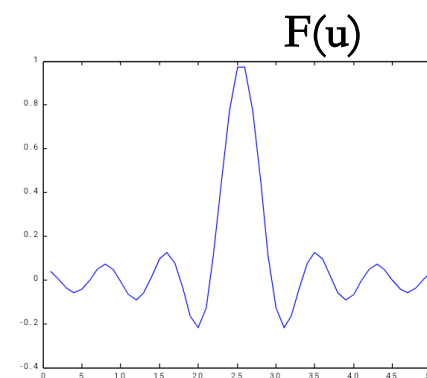
函数

$$\Lambda(x) = \begin{cases} 1 - |x|, & |x| \leq 1; \\ 0, & \text{其它.} \end{cases}$$



变换

$$\sin^2(u)$$



通过减少透镜边缘的透射率，可实现超分辨

(三) 常用傅里叶变换式

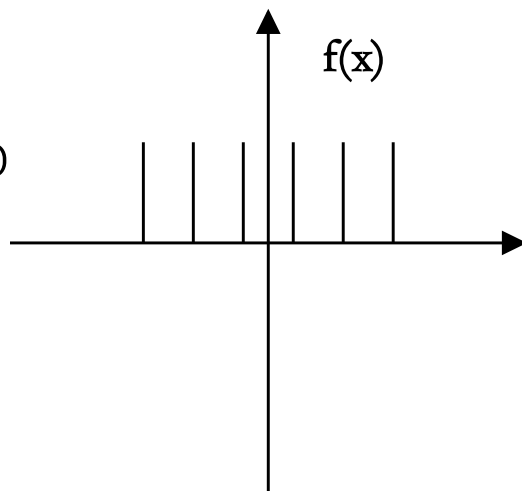
函数

变换

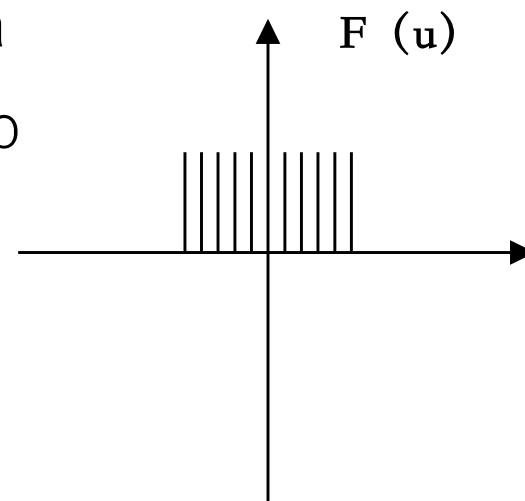
$$\text{comb}\left(\frac{x}{a}\right) = \sum_{-\infty}^{+\infty} \delta\left(\frac{x}{a} - n\right) \quad a \cdot \text{comb}(au) = a \sum_{-\infty}^{+\infty} \delta(au - n)$$

梳状函数

{ 栅缝无限窄
间距 a
宽度 ∞



{ 谱线无限窄
间距 $\frac{1}{a}$
宽度 ∞



(三) 常用傅里叶变换式

函数

$$\text{Sgn}(x) \cdot \text{Sgn}(y)$$

$$\text{Sgn} = \begin{cases} 1, & x > 0 \\ 0, & x = 0 \\ -1, & x < 0 \end{cases}$$

变换

$$\left(\frac{1}{j\pi u}\right) \cdot \left(\frac{1}{j\pi v}\right)$$

一个位相函数的变换，模值反比于谱坐标。

(三) 常用傅里叶变换式

函数

$$\frac{\partial [g(x, y)]}{\partial x}$$

变换

$$j2\pi u G(u, v)$$

微分定理，像增加中常用，加强信号对比，减少背景水平。

(四) 关于 δ 函数的补充

1、定义：
$$\delta(x, y) = \begin{cases} 0 & \text{otherwise} \\ \infty & x = y = 0 \end{cases}$$

2、性质：
$$\delta(ax) = \frac{1}{a} \delta(x), \quad \delta(x - a) = \delta(a - x)$$

3、卷积：
$$\int \delta(x - a) f(x) dx = f(a)$$

4、平移：
$$f(x - a) = f(x) * \delta(x - a)$$

5、
$$F\{\delta(x)\} = 1$$

二 透镜的F.T性质

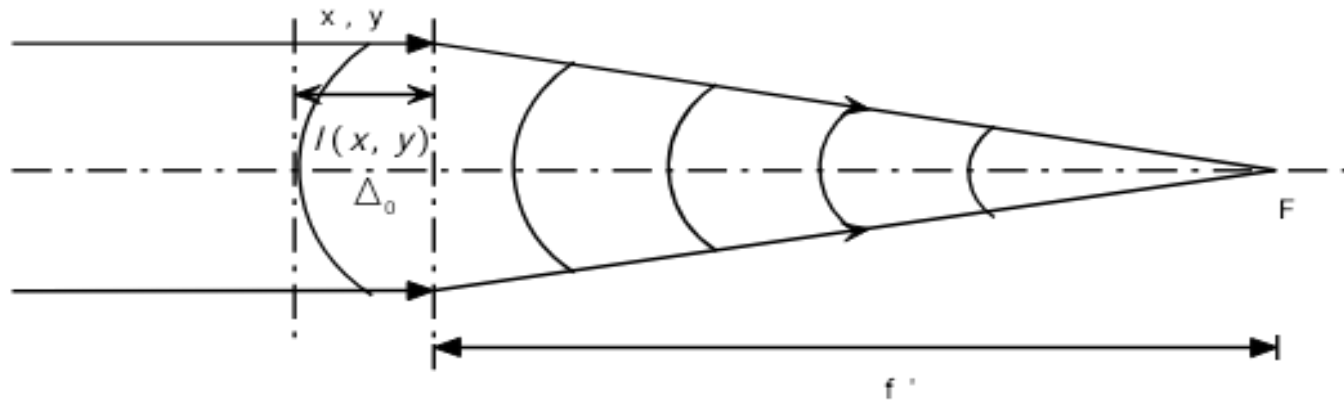
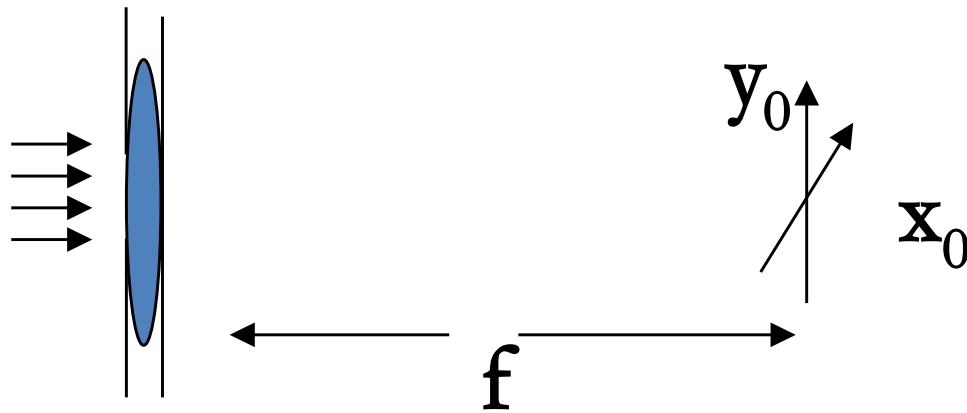


图 3.1 薄透镜的位相变换作用

$$U'_l(x, y) = t_l(x, y)U_l(x, y)$$

$$t_l(x, y) = e^{jkl}$$

平面波经过正透镜后会变成会聚的球面波。



薄透镜

$$\frac{1}{f} = (n-1) \left(\frac{1}{R_1} - \frac{1}{R_2} \right)$$

根据费马原理

$$n\Delta_0 + f = l(x, y) + \sqrt{f^2 + (x^2 + y^2)}$$

由傍轴近似

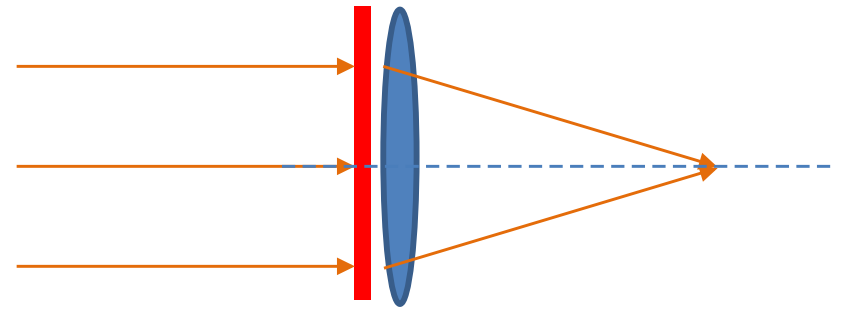
$$l(x, y) = n\Delta_0 - (x^2 + y^2) / 2f$$

$$t_l(x, y) = e^{jkn\Delta_0} e^{-jk(x_1^2 + y_1^2)/2f}$$

常数位相延迟

与球面波相联系的二次位相因子

物体紧贴透镜放置



若物体透射率函数为 $t_0(x, y)$ ，则输入分布函数为

$$U_i(x_1, y_1) = A t_0(x, y)$$

考虑透镜的孔径是有限的，则透镜输出面上的复振幅分布为

$$U'_i(x_1, y_1) = A t_0(x, y) P(x, y) \exp\left[-i \frac{k}{2f} (x^2 + y^2)\right]$$

由菲涅尔衍射积分公式可得透镜后焦平面的分布为

$$U_f(x_f, y_f) = \frac{\exp(ikf)}{j\lambda f} \exp\left[i \frac{k}{2f} (x_f^2 + y_f^2)\right] \cdot \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \{ A t_0(x, y) P(x, y) \cdot \exp\left[-i \frac{2\pi}{\lambda f} (x_f x_1 + y_f y_1)\right] dx_1 dy_1$$

$$U_f(x_f, y_f) = \frac{\exp(ikf)}{i\lambda f} \exp\left[i\frac{k}{2f}(x_f^2 + y_f^2)\right] \cdot$$

$$\cdot \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \{ A t_0(x, y) P(x, y) \cdot \exp\left[-i\frac{2\pi}{\lambda f}(x_f x_1 + y_f y_1)\right] dx_1 dy_1$$

不是严格的傅里叶变换，有相位弯曲

空间频率 $\frac{x_f}{\lambda f}, \frac{y_f}{\lambda f}$

物体紧贴透镜后面，分布一样

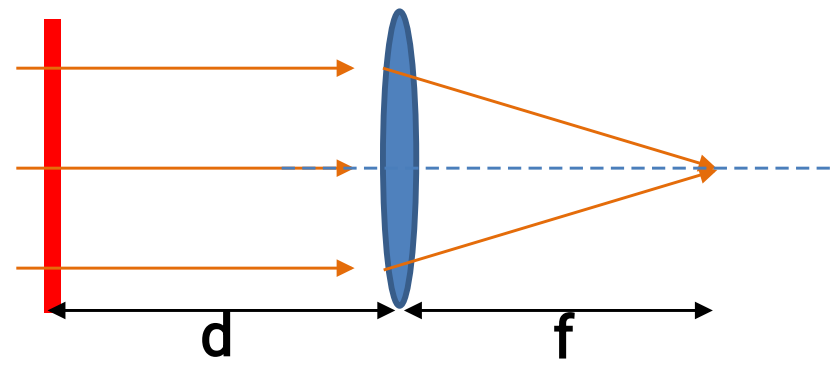
强度分布

$$I_f(x_f, y_f) = \frac{A^2}{\lambda^2 f^2} \left| \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \{ t_0(x, y) P(x, y) \cdot \exp\left[-i\frac{2\pi}{\lambda f}(x_f x_1 + y_f y_1)\right] dx_1 dy_1 \right|^2$$

若物体比透镜小，则光瞳函数？

物体放在透镜前方d处

透镜输入面处的频谱分布函数为



$$F\{U_i(x_i, y_i)\} = \exp(ikd) \exp[-i\pi\lambda d(f_x^2 + f_y^2)] F\{At_0(x_0, y_0)\}$$

于是

$$U_f(x_f, y_f) = \frac{\exp(ikf)}{i\lambda f} \exp[i\frac{k}{2f}(x_f^2 + y_f^2)] \cdot F\{U_i(x, y)\}$$

$$U_f(x_f, y_f) = \frac{\exp(ikf)}{i\lambda f} \exp[i\frac{k}{2f}(x_f^2 + y_f^2)] \cdot \exp(ikd) \exp[-i\pi\lambda d(f_x^2 + f_y^2)] F\{At_0(x_0, y_0)\}$$

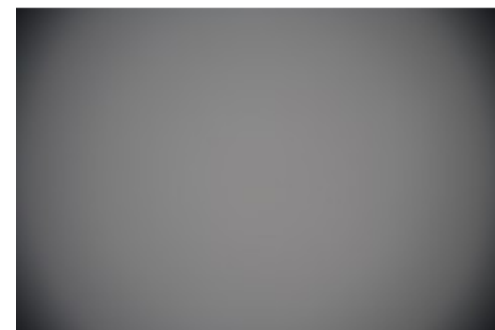
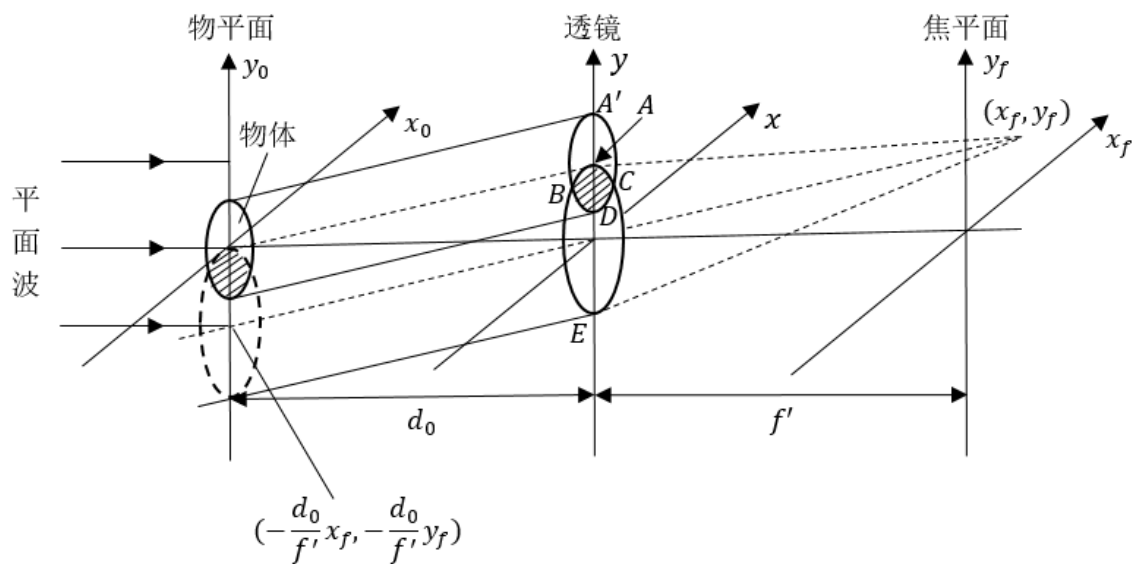
$$= \frac{A \exp(ikf)}{i\lambda f} \exp[i\frac{k(1-d/f)}{2f}(x_f^2 + y_f^2)] \cdot \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \{t_0(x_0, y_0) \cdot \exp[-i\frac{2\pi}{\lambda f}(x_f x_0 + y_f y_0)] dx_0 dy_0\}$$

$$f_x = \frac{x_f}{\lambda f}, \quad f_y = \frac{y_f}{\lambda f}$$

若 $d = f$

$$U_f(x_f, y_f) = \frac{A \exp(ikf)}{i\lambda f} \int \int_{-\infty}^{\infty} \{t_0(x_0, y_0) \cdot \exp[-i \frac{2\pi}{\lambda f} (x_f x_0 + y_f y_0)]\} dx_0 dy_0$$

渐晕效应

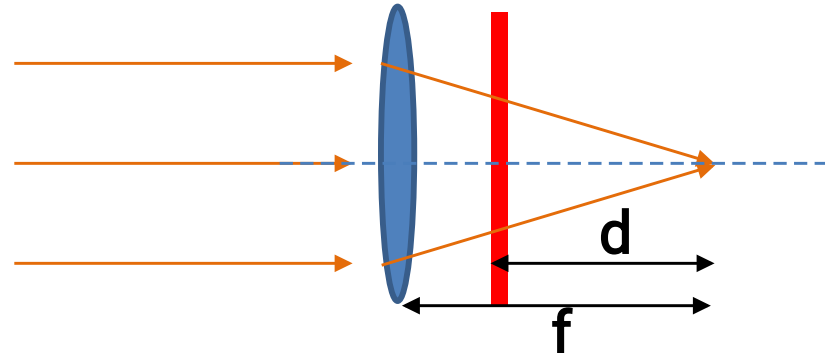


$$U_f(x_f, y_f) = \frac{A \exp(ikf')}{i\lambda f'} \exp[i \frac{k(1-d_0/f')}{2f'} (x_f^2 + y_f^2)] .$$

$$\int \int_{-\infty}^{\infty} \{ t_0(x_0, y_0) P(x_0 + \frac{d_0}{f'} x_f, y_0 + \frac{d_0}{f'} y_f) \cdot \exp[-i \frac{2\pi}{\lambda f'} (x_f x_0 + y_f y_0)] dx_0 dy_0$$

物体放在透镜后方f-d处

经透镜后变为汇聚的球面波



$$A \exp\left[-i \frac{k}{2f} (x^2 + y^2)\right]$$

到达物体时，球面波为

$$\frac{A f}{d} \exp\left[-i \frac{k}{2d} (x_0^2 + y_0^2)\right]$$

物体被照明区域的直径为

$$l' = \frac{d}{f} l$$

l 为透镜直径

因此透过物体的光振幅分布为

$$U_0(x_0, y_0) = \frac{A f'}{d} \underbrace{P\left(x_0 \frac{f}{d}, y_0 \frac{f}{d}\right)}_{\text{有效光瞳函数}} \exp\left[-i \frac{k}{2d} (x_0^2 + y_0^2)\right] \cdot t_0(x_0, y_0)$$

有效光瞳函数

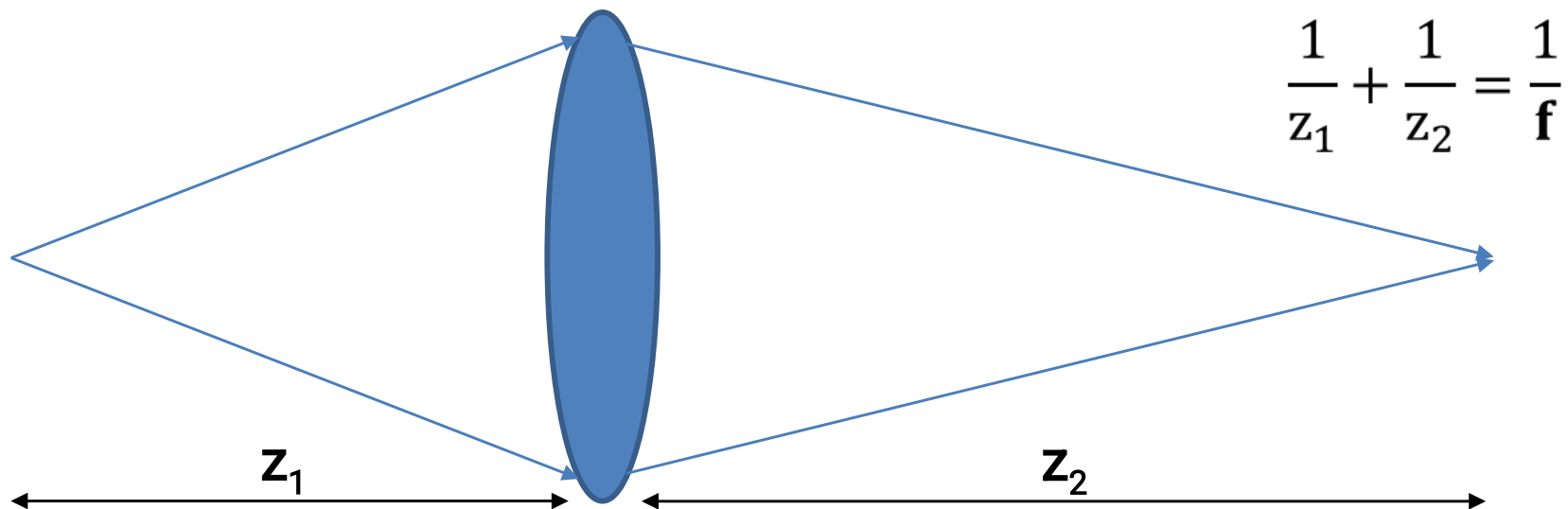
物体到频谱面为菲涅尔衍射，于是

$$U_f(x_f, y_f) = \frac{A \exp(ikd)}{i\lambda d} \frac{f}{d} \exp\left[i \frac{k}{2d} (x_f^2 + y_f^2)\right] \cdot \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \left\{ t_0(x_0, y_0) P\left(x_0 \frac{f}{d}, y_0 \frac{f}{d}\right) \cdot \exp\left[-i \frac{2\pi}{\lambda d} (x_f x_0 + y_f y_0)\right] \right\} dx_0 dy_0$$

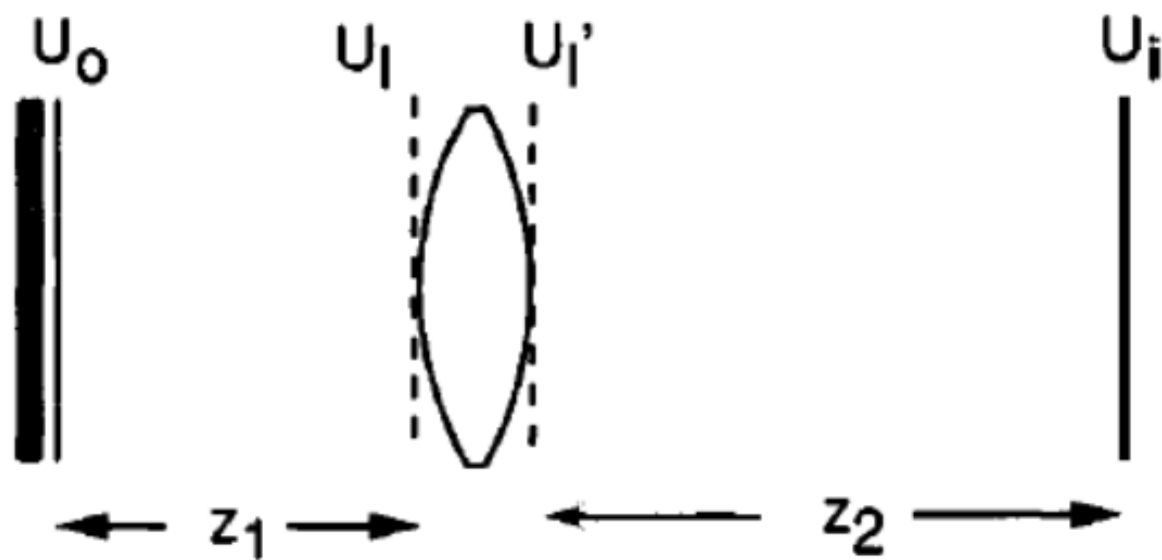
具有改变空间频标的灵活性，d增大，频标 $f_x = \frac{x_f}{\lambda d}$, $f_y = \frac{y_f}{\lambda d}$ 变小，傅里叶变换的空间分布范围变大。

三、透镜成像的傅里叶分析

几何光学

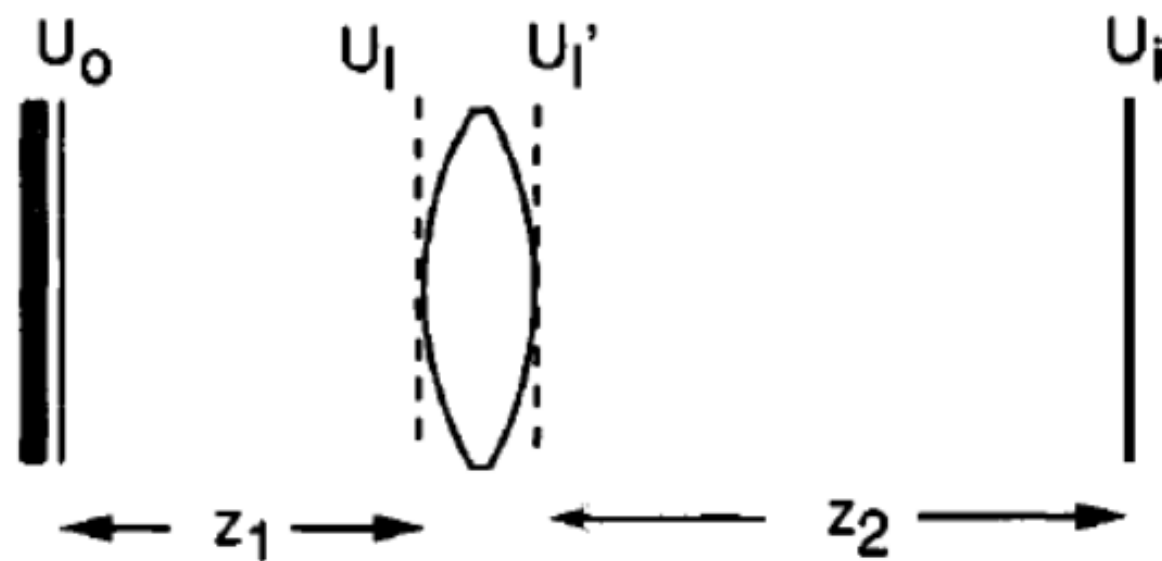


傅里叶光学



$$U_i(u, v) = \iint_{-\infty}^{\infty} h(u, v; \xi, \eta) U_o(\xi, \eta) d\xi d\eta$$

脉冲函数的推导

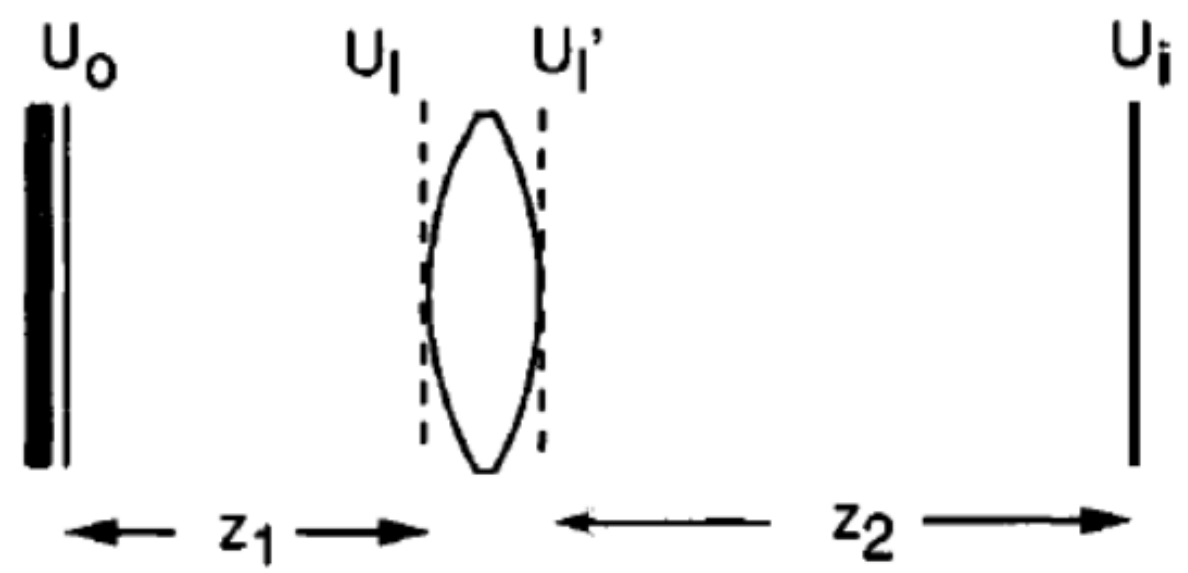


$$U_l(x, y) = \frac{1}{j\lambda z_1} \exp \left\{ j \frac{k}{2z_1} [(x - \xi)^2 + (y - \eta)^2] \right\}.$$

$$U_l'(x, y) = U_l(x, y) P(x, y) \exp \left[-j \frac{k}{2f} (x^2 + y^2) \right]$$

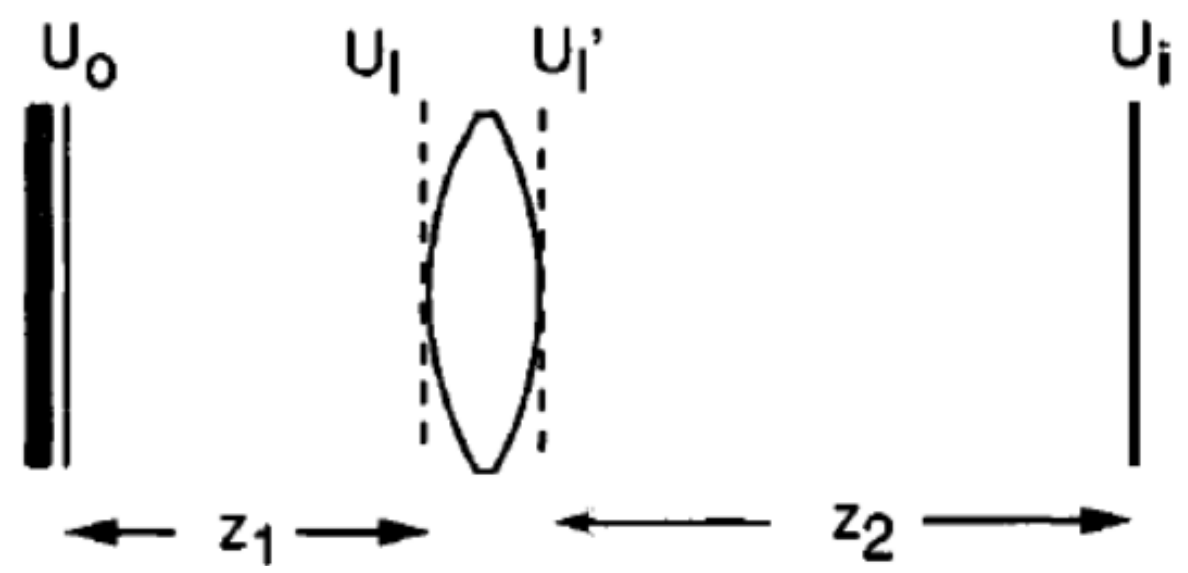
$$h(u, v; \xi, \eta) = \frac{1}{j\lambda z_2} \iint_{-\infty}^{\infty} U_l'(x, y) \exp \left\{ j \frac{k}{2z_2} [(u - x)^2 + (v - y)^2] \right\} dx dy$$

脉冲函数的推导



$$\begin{aligned} h(u, v; \xi, \eta) = & \frac{1}{\lambda^2 z_1 z_2} \exp \left[j \frac{k}{2 z_2} (u^2 + v^2) \right] \exp \left[j \frac{k}{2 z_1} (\xi^2 + \eta^2) \right] \\ & \times \iint_{-\infty}^{\infty} P(x, y) \exp \left[j \frac{k}{2} \left(\frac{1}{z_1} + \frac{1}{z_2} - \frac{1}{f} \right) (x^2 + y^2) \right] \\ & \times \exp \left\{ -jk \left[\left(\frac{\xi}{z_1} + \frac{u}{z_2} \right) x + \left(\frac{\eta}{z_1} + \frac{v}{z_2} \right) y \right] \right\} dx dy. \end{aligned}$$

脉冲函数的推导



$$h(u, v; \xi, \eta) \approx \frac{1}{\lambda^2 z_1 z_2} \iint_{-\infty}^{\infty} P(x, y) \times \exp \left\{ -jk \left[\left(\frac{\xi}{z_1} + \frac{u}{z_2} \right) x + \left(\frac{\eta}{z_1} + \frac{v}{z_2} \right) y \right] \right\} dx dy.$$

$$M = -\frac{z_2}{z_1} \quad h(u, v; \xi, \eta) \approx \frac{1}{\lambda^2 z_1 z_2} \iint_{-\infty}^{\infty} P(x, y) \times \exp \left\{ -j \frac{2\pi}{\lambda z_2} [(u - M\xi)x + (v - M\eta)y] \right\} dx dy.$$

如果P为半径是R的圆孔，则 $h(u, v; \xi, \eta) = \frac{J_1(k \frac{R}{z_2} r)}{k \frac{R}{z_2} r} \quad r^2 = (u - M\xi)^2 + (v - M\eta)^2$

物像关系

$$h(u, v; \tilde{\xi}, \tilde{\eta}) = \frac{1}{\lambda^2 z_1 z_2} \iint_{-\infty}^{\infty} P(x, y) \times \exp \left\{ -j \frac{2\pi}{\lambda z_2} \left[(u - \tilde{\xi})x + (v - \tilde{\eta})y \right] \right\} dx dy$$

$$\tilde{\xi} = M\xi \quad \tilde{\eta} = M\eta$$

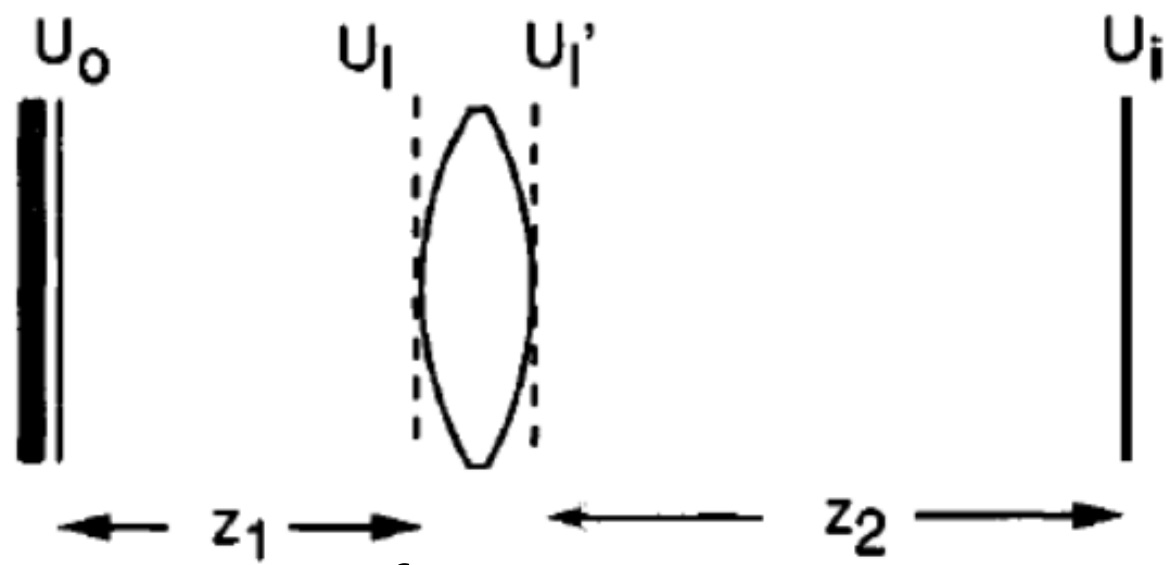
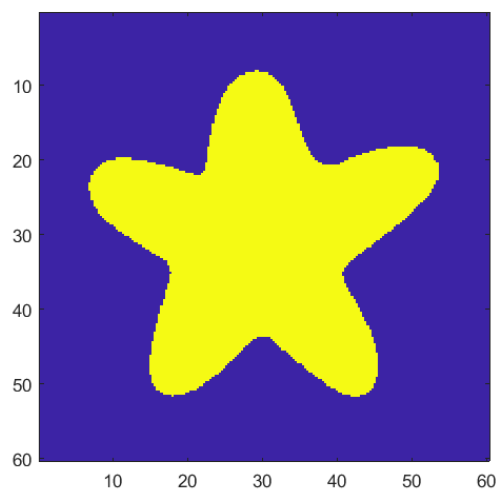
$$\tilde{x} = \frac{x}{\lambda z_2} \quad \tilde{y} = \frac{y}{\lambda z_2} \quad \tilde{h} = \frac{1}{|M|} h.$$

$$U_i(u, v) = \iint_{-\infty}^{\infty} \tilde{h}(u - \tilde{\xi}, v - \tilde{\eta}) \left[\frac{1}{|M|} U_o \left(\frac{\tilde{\xi}}{M}, \frac{\tilde{\eta}}{M} \right) \right] d\tilde{\xi} d\tilde{\eta},$$

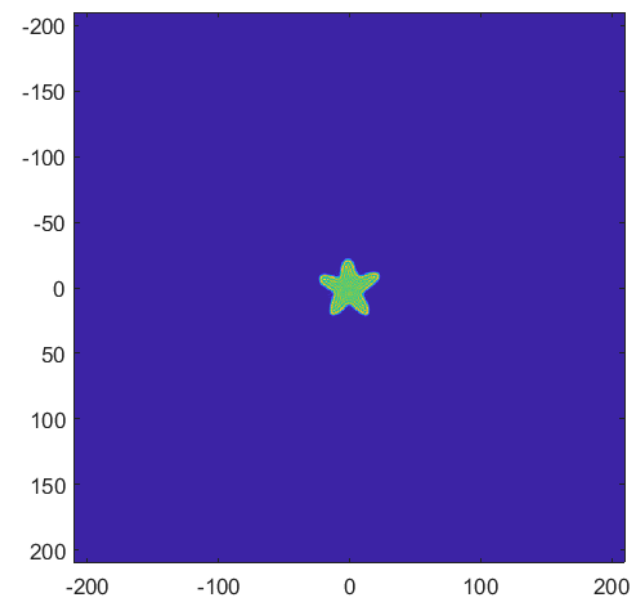
$$U_i(u, v) = \tilde{h}(u, v) \otimes U_g(u, v)$$

$$U_g(u, v) = \frac{1}{|M|} U_o \left(\frac{u}{M}, \frac{v}{M} \right)$$

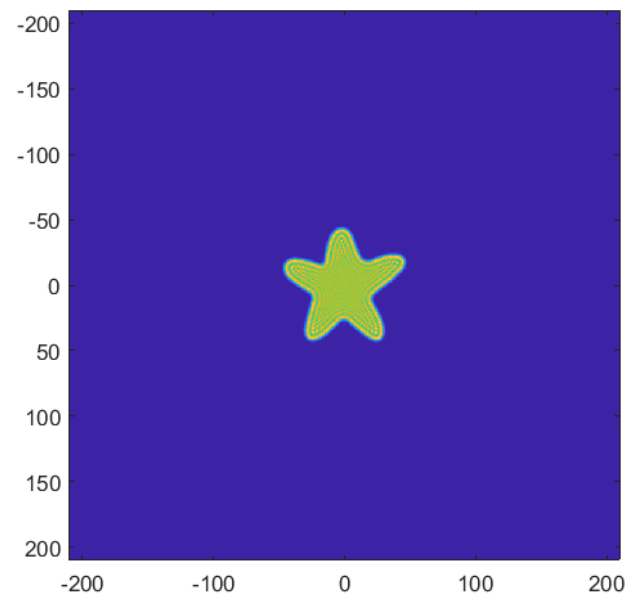
成像的数值模拟：



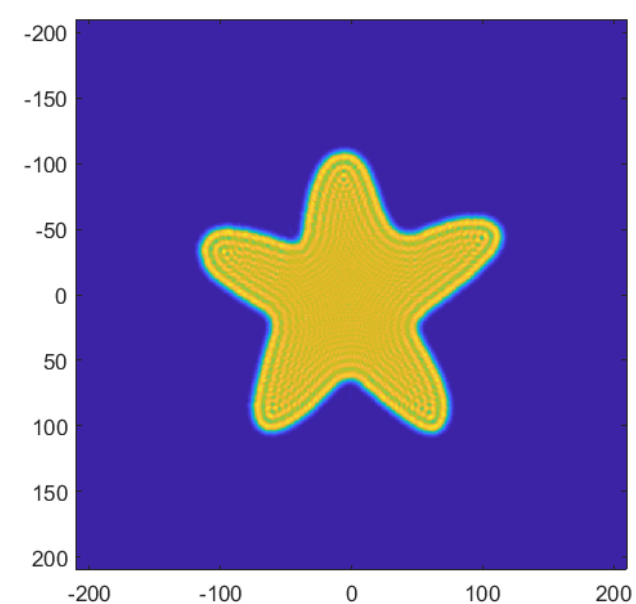
$f=100\mu\text{m}$
 $D=420\mu\text{m}$
Wavelength= $0.633\mu\text{m}$



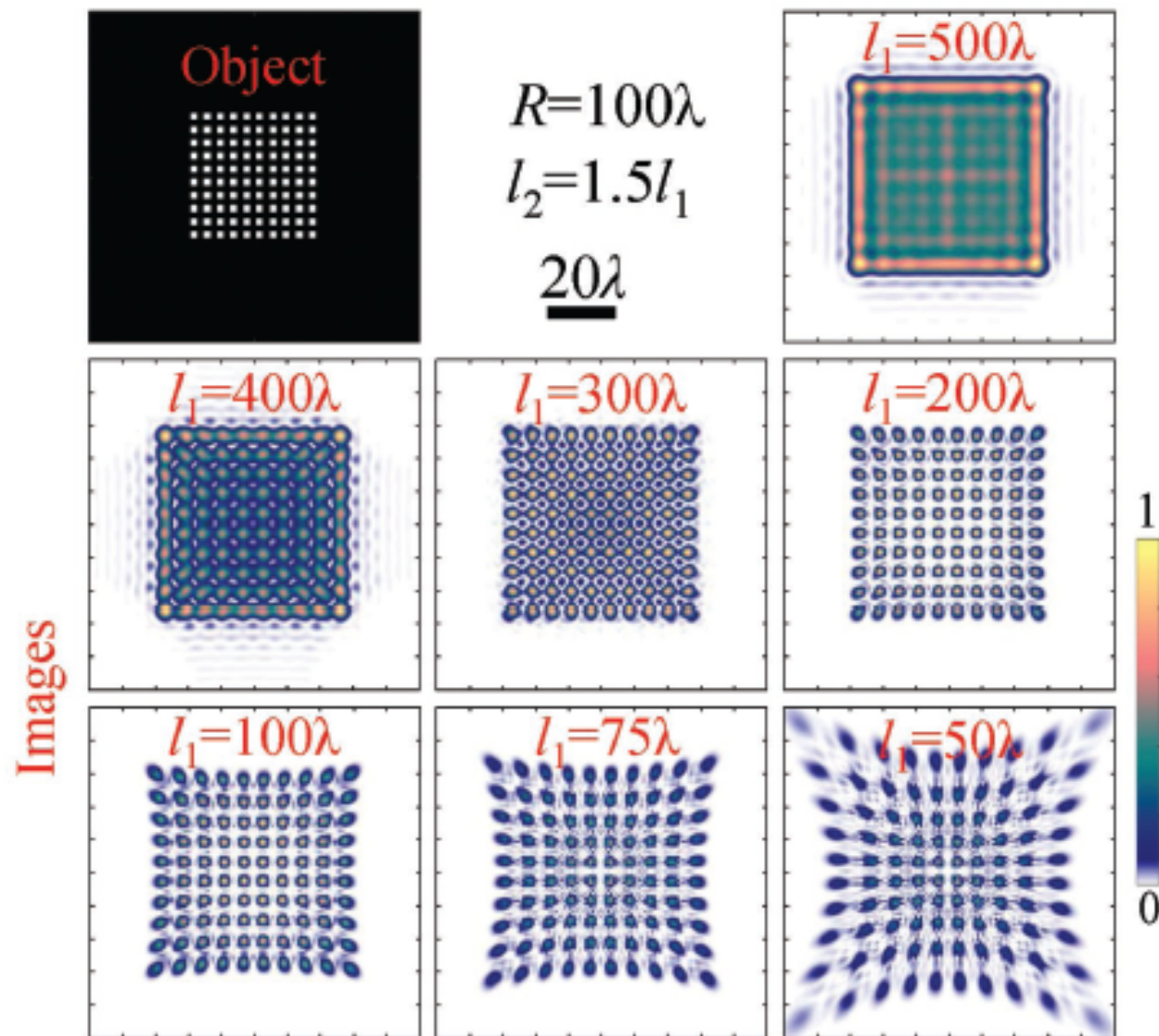
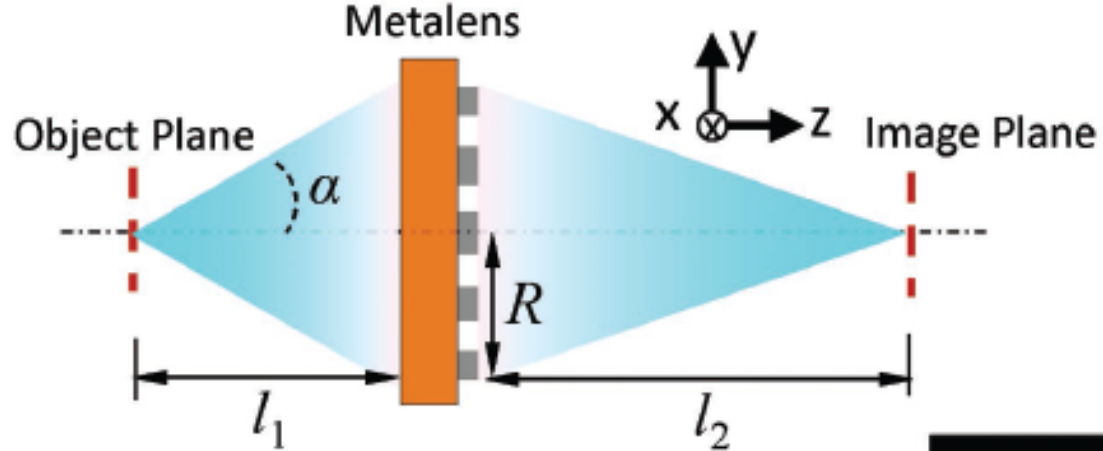
$M=1$



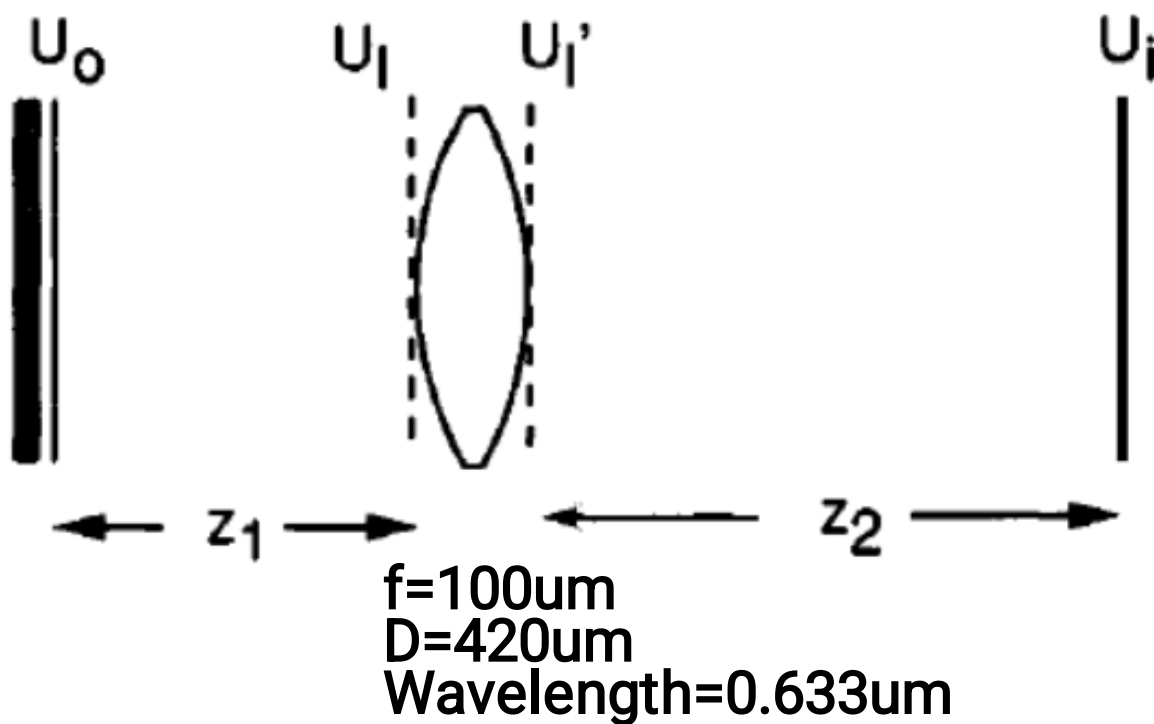
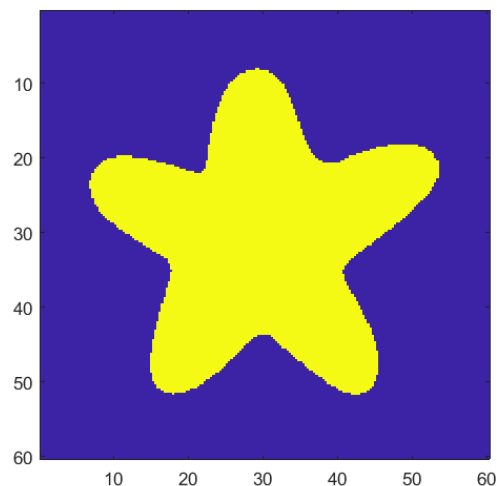
$M=2$



$M=6$



成像的数值模拟作业：



利用角谱方法或RS衍射积分，采用光场逐层传播的方式（即第一次传播 z_1 ，所得光场乘以透镜位相，然后进行第二次传播 z_2 ），数值模拟的物体在不同放大倍率下的像： $M=1$ ； $M=2$ ， $M=6$.具体成像结果可参考上页slide。

四、夫朗和费衍射光场的计算

夫朗和费衍射光场的计算

1、确定透过率函数： $t(x_1, y_1)$

2、在单位振幅、单色、平面波、垂直照明的情况下： $U(x_1, y_1) = t(x_1, y_1)$

3、作傅立叶变换： $F\{U(x_1, y_1)\}$

4、 $U(x_0, y_0) = \frac{e^{jkz}}{j\lambda z} e^{j\frac{k}{2z}(x_0^2 + y_0^2)} F\{U(x_1, y_1)\}$

5、 $I(x_0, y_0) = U(x_0, y_0)U^*(x_0, y_0)$

例1 单色平面波垂直照明，距离d，狭缝宽度极小两衍射屏

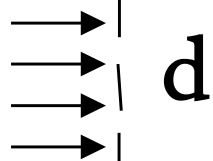
$$t(x_1) = \delta\left(x - \frac{d}{2}\right) + \delta\left(x + \frac{d}{2}\right)$$

$$F\{t(x_1)\} = e^{-j\pi u d} + e^{j\pi u d} = 2 \cos \pi u d$$

$$U(x_0) = c F\{U(x_1)\}$$

$$I(x_0) = I_0 \cos^2 \pi u d = I_0 (1 - \cos 2\pi u d) / 2$$

$(\mathbf{x}_1, y_1)$



d

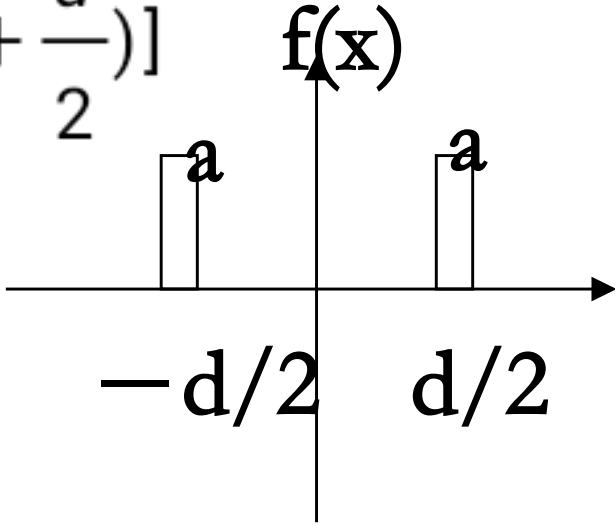
$(\mathbf{x}_0, y_0)$

$$u = x_0 / z\lambda$$

例2 双缝衍射

$$t(x_1) = \text{rect}\left(\frac{x_1}{a} - \frac{d}{2a}\right) + \text{rect}\left(\frac{x_1}{a} + \frac{d}{2a}\right)$$

$$= \text{rect}\left(\frac{x}{a}\right) * \left[\delta\left(x - \frac{d}{2}\right) + \delta\left(x + \frac{d}{2}\right) \right]$$

$$F\{t(x_1)\} = 2a \sin c(au) \cos 2\pi u \frac{d}{2}$$


$$I(x_0, y_0) = I_0 \sin^2(au) \cos^2 \pi u d$$

$$= \frac{2a^2}{z^2 \lambda^2} \sin^2 \frac{ax_0}{\lambda z} \left(1 + \cos 2\pi \frac{dx_0}{\lambda z}\right)$$

$$u = \frac{\cos \alpha}{\lambda} = \frac{x_0}{\lambda z}$$

例3 三缝衍射

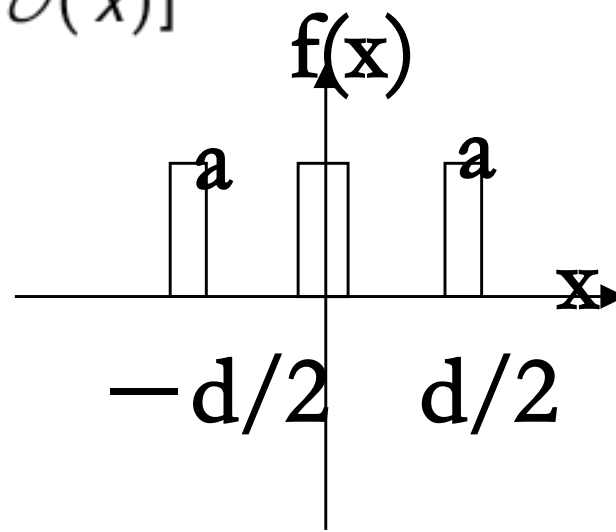
$$t(x) = \text{rect}\left(\frac{x}{a} - \frac{d}{2a}\right) + \text{rect}\left(\frac{x}{a} + \frac{d}{2a}\right) + \text{rect}\left(\frac{x}{a}\right)$$

$$= \text{rect}\left(\frac{x}{a}\right) * \left[\delta\left(x - \frac{d}{2}\right) + \delta\left(x + \frac{d}{2}\right) + \delta(x) \right]$$

$$F\{t(x)\} = a \sin c(au) \left[1 + 2 \cos 2\pi u \frac{d}{2} \right]$$

$$I(x_0, y_0) = I_0 \sin^2(au) \cos^2 \pi u d$$

$$= \frac{a^2}{z^2 \lambda^2} \sin^2 \frac{ax_0}{\lambda z} \left(1 + 2 \cos 2\pi \frac{dx_0}{\lambda z} \right)^2$$



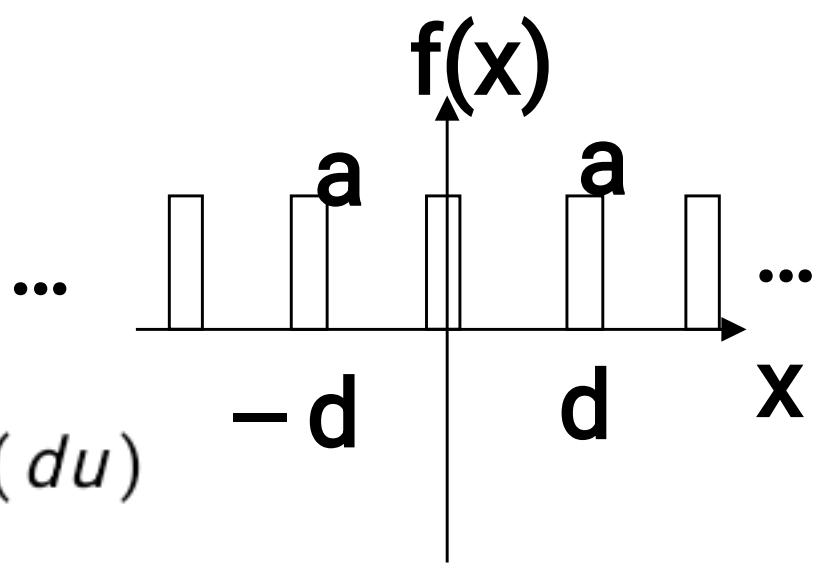
例4 光栅衍射（缝无限）

$$t(x_1) = rect(\frac{x_1}{a}) * comb(\frac{x_1}{d})$$

$$F\{t(x_1)\} = a \sin c(au) d \cdot comb(du)$$

$$U(x_0, y_0) = \frac{e^{jkz} \cdot e^{\frac{jk}{2z}(x_0^2 + y_0^2)}}{j\lambda z} ad \sin c(au) \cdot comb(du)$$

$$I(x_0, y_0) = \frac{a^2 d^2}{\lambda^2 z^2} \sin^2 c(au) \cdot comb^2(du)$$



例5 光栅衍射（N个缝）

$$t(x_1) = \text{rect}\left(\frac{x_1}{a}\right) * \sum \delta(x - nd)$$

$$F\{t(x_1)\} = a \sin c(au) \frac{1 - e^{-i2N\pi ud}}{1 - e^{-i2\pi ud}}$$

$$= a \sin c(au) \frac{2ie^{-iN\pi ud} \sin(\pi Nud)}{2ie^{-i\pi ud} \sin(\pi ud)}$$

$$I(x_0, y_0) = \frac{a^2}{\lambda^2 z^2} \sin^2 c(au) \cdot \frac{\sin^2(\pi Nud)}{\sin^2(\pi ud)}$$

单缝衍射因子 多光束干涉因子

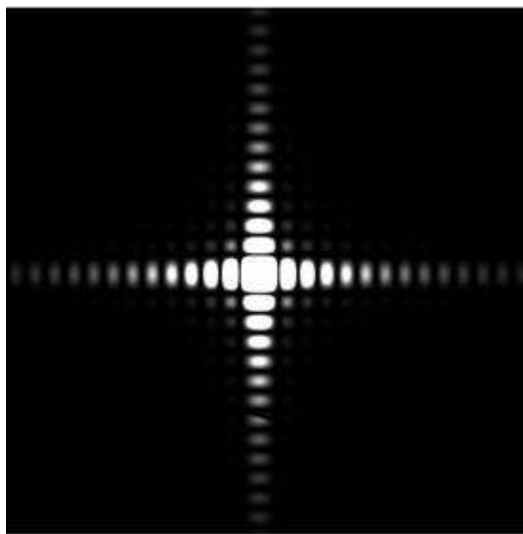
例6 矩形孔径

透过率函数为 $t(x_1, y_1) = \text{rect}\left(\frac{x_1}{l_x}\right) \text{rect}\left(\frac{y_1}{l_y}\right)$

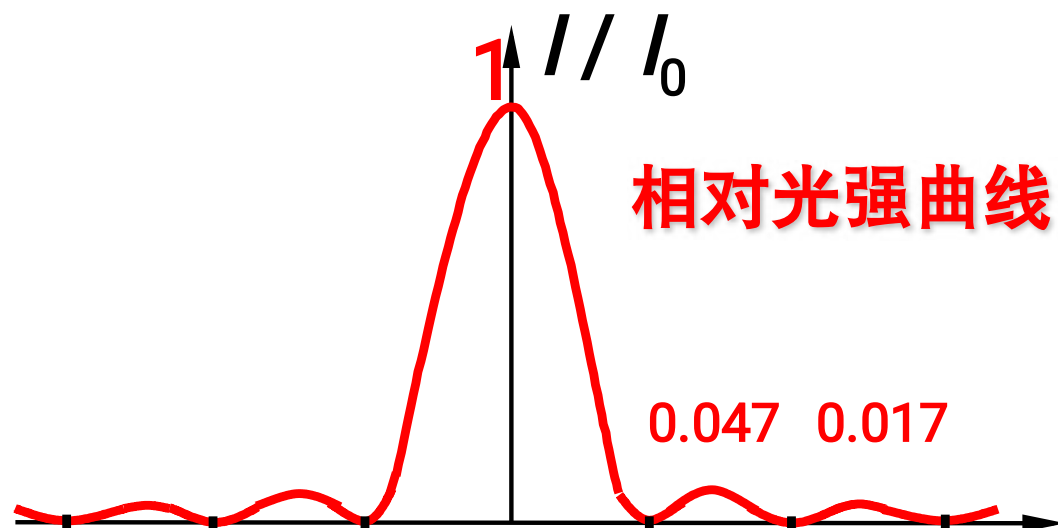
夫朗和费衍射振幅和强度分布分别为

$$U(x_0, y_0) = \frac{\exp(jkz)}{j\lambda z} \exp\left[j\frac{k}{2z}(x_0^2 + y_0^2)\right] l_x l_y \text{sinc}\left(\frac{l_x x_0}{\lambda z}\right) \text{sinc}\left(\frac{l_y y_0}{\lambda z}\right)$$

$$I(x_0, y_0) = \frac{l_x^2 l_y^2}{\lambda^2 z^2} \text{sinc}^2\left(\frac{l_x x_0}{\lambda z}\right) \text{sinc}^2\left(\frac{l_y y_0}{\lambda z}\right)$$



矩形孔径



截面图

夫朗和费衍射图样的例子

例7 圆形孔径

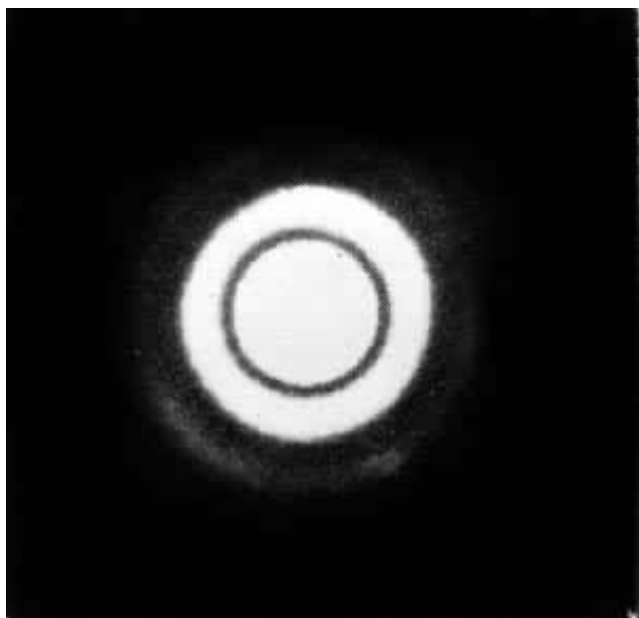
透过率函数为
$$t(r_1) = \text{circ} \left(\frac{r_1}{l/2} \right)$$

它的夫朗和费衍射振幅和强度分布分别为

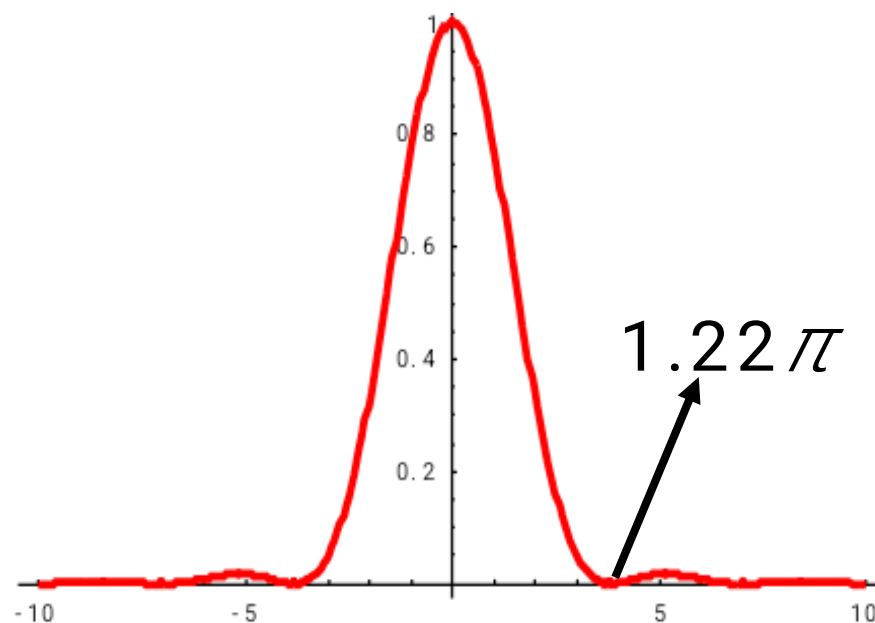
$$U(r_0) = \exp(jkz) \exp\left(j \frac{k}{2z} r_0^2\right) \frac{kl^2}{j8z} \left[2 \frac{J_1\left(\frac{klr_0}{2z}\right)}{\frac{klr_0}{2z}} \right]$$

$$I(r_0) = \left(\frac{kl^2}{8z}\right)^2 \left[2 \frac{J_1\left(\frac{klr_0}{2z}\right)}{\frac{klr_0}{2z}} \right]^2$$

夫朗和费衍射图样的例子



圆形孔径



截面图

$$\Delta\theta = 0.61 \frac{\lambda}{a} = 1.22 \frac{\lambda}{l}$$

夫朗和费衍射图样的例子

环形孔径

透过率函数为 $t(r_1) = \text{circ}(\frac{r_1}{l/2}) - \text{circ}(\frac{r_1}{\varepsilon l/2})$

夫朗和费衍射图样的振幅分布为

$$U(r_0) = \exp(jkz) \exp(j \frac{k}{2z} r_0^2) \frac{kl^2}{j8z} \left[2 \frac{J_1(\frac{klr_0}{2z})}{\frac{klr_0}{2z}} - \varepsilon^2 2 \frac{J_1(\frac{k\varepsilon l r_0}{2z})}{\frac{k\varepsilon l r_0}{2z}} \right]$$

强度分布为

$$I(r_0) = \frac{1}{(1 - \varepsilon^2)^2} \left[2 \frac{J_1(\frac{klr_0}{2z})}{\frac{klr_0}{2z}} - \varepsilon^2 2 \frac{J_1(\frac{k\varepsilon l r_0}{2z})}{\frac{k\varepsilon l r_0}{2z}} \right]^2 I_0$$

其中衍射图样中心的光强

$$I_0 = \left(\frac{kl^2}{8z} \right)^2 (1 - \varepsilon^2)^2$$

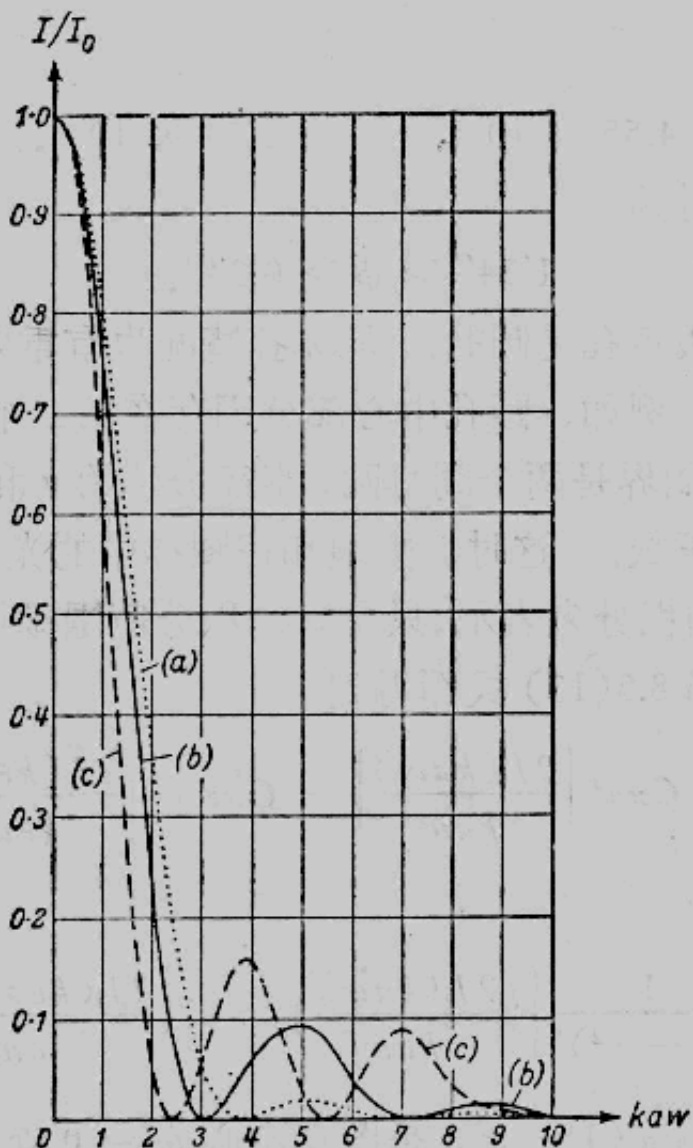


图 8.30 中心遮挡对分辨的影响图示. 夫琅和费图样归一化强度曲线

(a) 圆孔, (b) 环孔, $\varepsilon = \frac{1}{2}$, (c) 环孔, $\varepsilon \rightarrow 1$

[引自, G. C. Steward, *The Symmetrical Optical System* (Cambridge University Press, 1928), p. 89.]

可见, 用环形孔径遮挡孔的中心部分时, 分辨本领将得到提高。

但同时, 象的亮度下降, 次极大 (旁瓣) 变得比较显著, 以致反差降低。

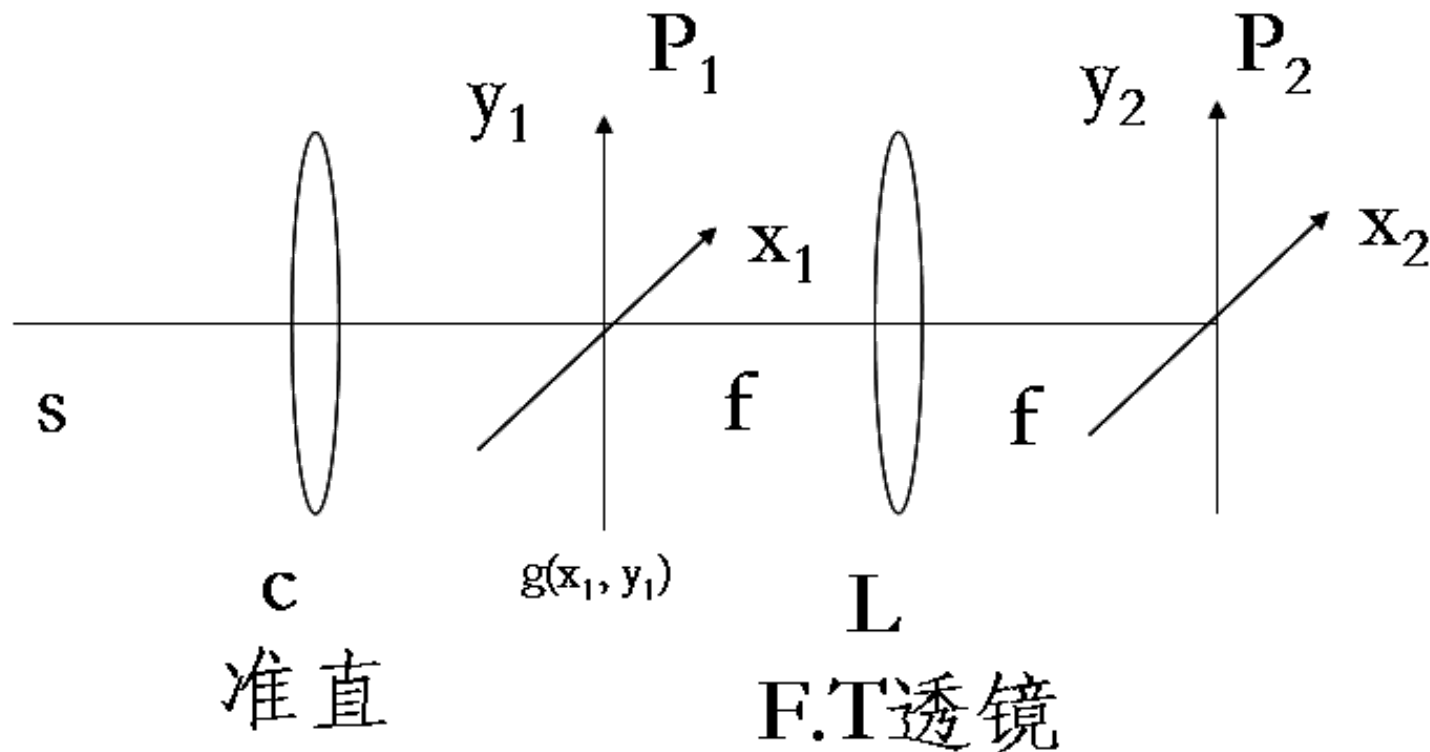
例如, $\varepsilon = 1/2$ 时, 中心光强度下降到 $9/16$ 。

第一次极大是主极大的10%, 而圆孔1.8%。

能量的损失可以通过使用位相型的环形孔径来避免

§3.5. 光学空间滤波 及全光图像处理

1、光学频谱分析处理：

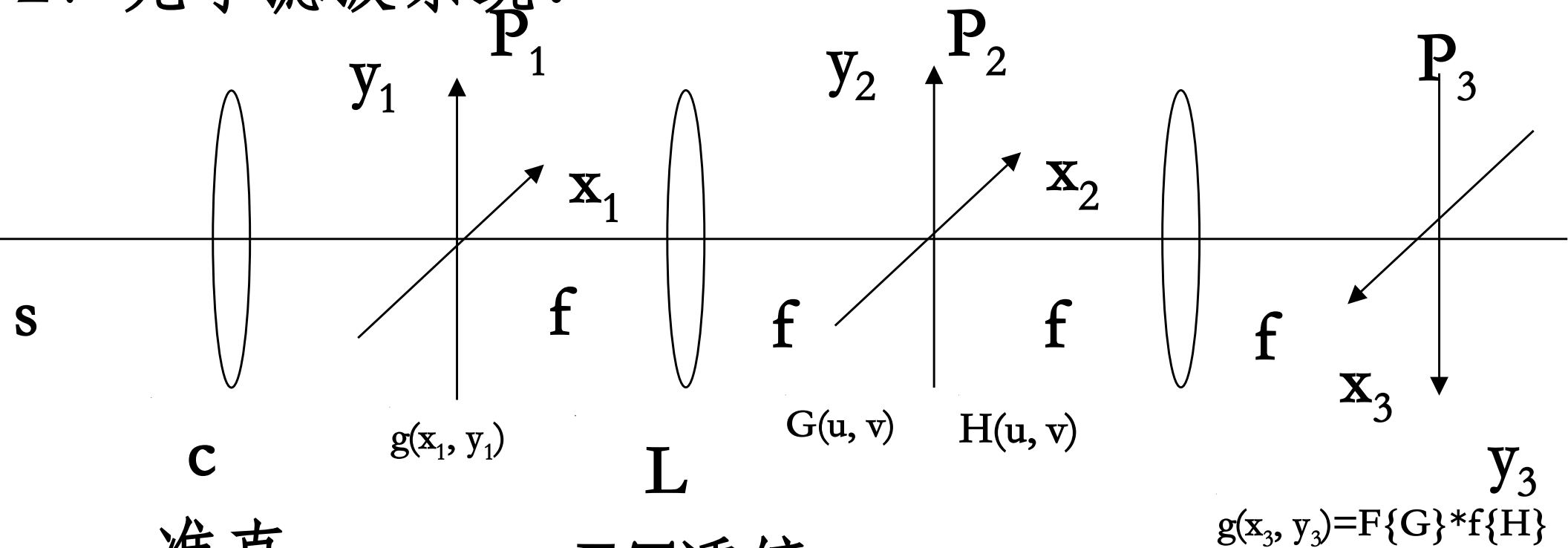


$g(x_1, y_1)$ 空间频谱分布：

$$G(u, v) = \iint_{-\infty}^{\infty} g(x_1, y_1) \exp[-i2\pi(ux_1 + vy_1)] dx_1 dy_1$$

$$G(x_2, y_2) = \iint_{-\infty}^{\infty} g(x_1, y_1) \exp[-i\frac{2\pi}{f\lambda}(x_2 x_1 + y_2 y_1)] dx_1 dy_1$$

2、光学滤波系统：



准直

F.T透镜

P₂ 上，g(x₁, y₁)空间频谱分布：

$$G(u, v) = \iint_{-\infty}^{\infty} g(x_1, y_1) \exp[-i2\pi(ux_1 + vy_1)] dx_1 dy_1$$

$$G(x_2, y_2) = \iint_{-\infty}^{\infty} g(x_1, y_1) \exp[-i\frac{2\pi}{f\lambda}(x_2 x_1 + y_2 y_1)] dx_1 dy_1$$

P_2 外:

$$U(x_2, y_2) = \frac{V_0}{f\lambda} G(u, v)$$

滤波片: $t_A(x_2, y_2) = KH(u, v)$

滤波后: $U_2(x_2, y_2) = \frac{V_0}{f\lambda} G(u, v) \cdot KH(u, v)$

$$U_3(x_3, y_3) = cF\{G(u, v)\} * F\{H(u, v)\}$$

3.5.2. 空间滤波器

空间滤波器

■ (一)、滤波器分类: $H(u, v) = A(u, v) e^{i\varphi(u, v)}$

1、振幅滤波器 $\varphi(u, v) = 0$, 按 $A(u, v)$ 制作

2、位相滤波器 $A(u, v)$ 常数, 改变 $\varphi(u, v)$

3、全息滤波器

4、计算滤波器

■ (二)、二元振幅全息滤波器 (0-1, 透光 - 不透光)

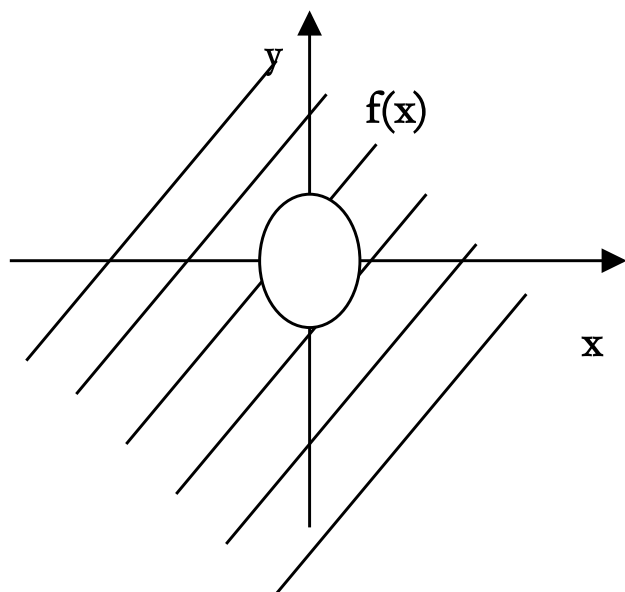
1、低通滤波:

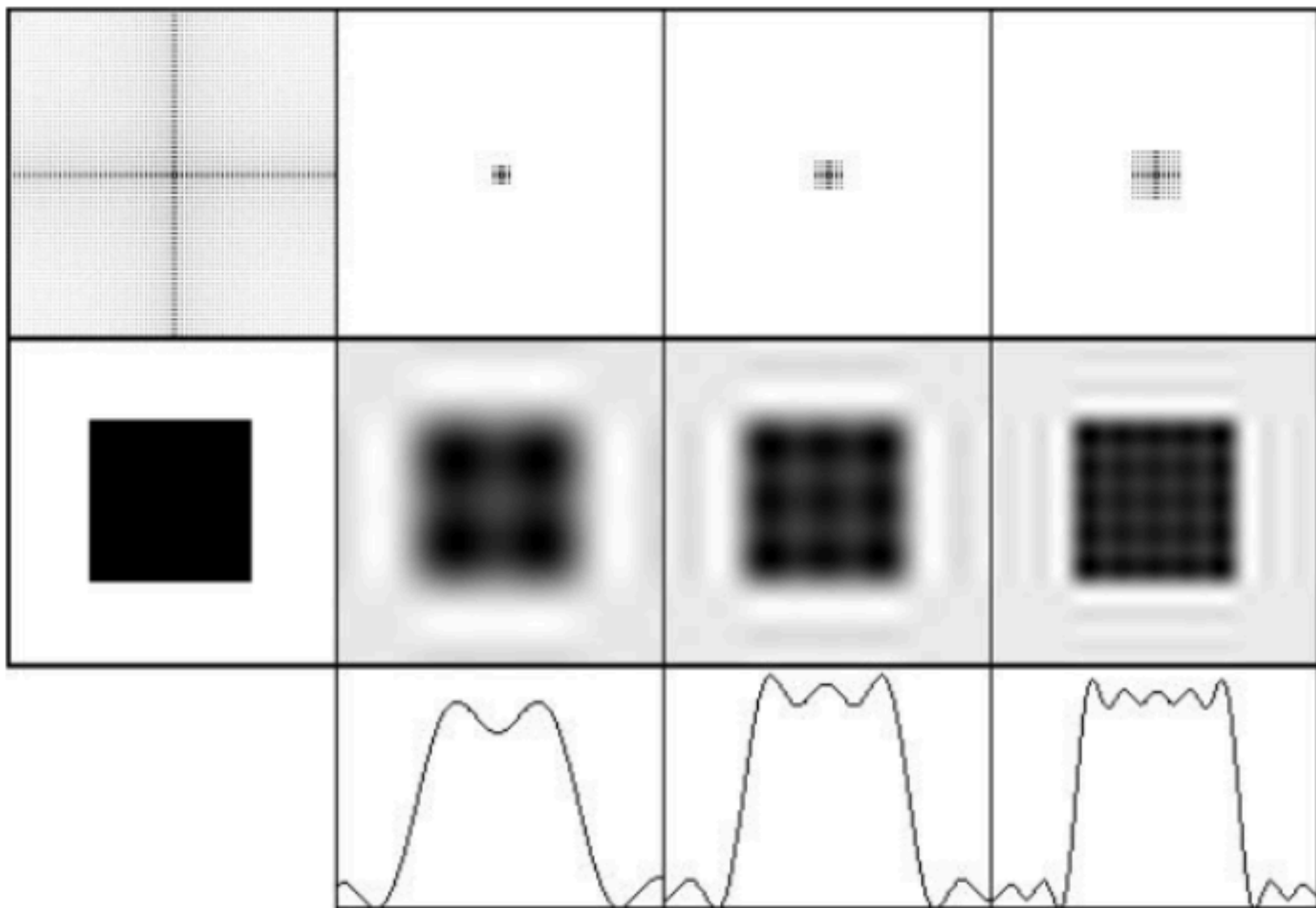
$$u = \frac{x_2}{\lambda f}, \quad v = \frac{y_2}{\lambda f}$$

空间频率:

$$\text{直径 } r = \sqrt{x_2^2 + y_2^2}$$

滤高频, 故可忽略细节,
可处理高反差或轮廓过
强的图像。





利用空间滤波提高光束质量

当光束质量不好时，如谐振腔或透镜引起的衍射，产生高频部分。可加一小孔——空间滤波，使第一透镜产生的爱里斑通过。

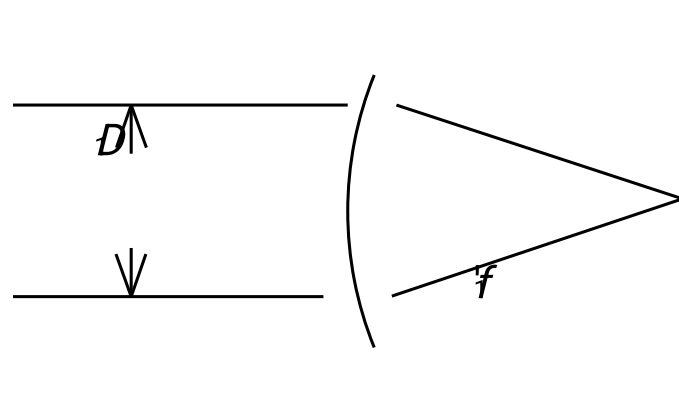
$$D_1 = 3\text{ mm}, f_1' = 5\text{ mm} = z$$

$$\Delta r_0 = 1.22 \frac{\lambda z}{D}$$

$$= 1.22 \times \frac{0.63 \times 10^{-3} \times 5}{3} = 1.3 \mu\text{m}$$

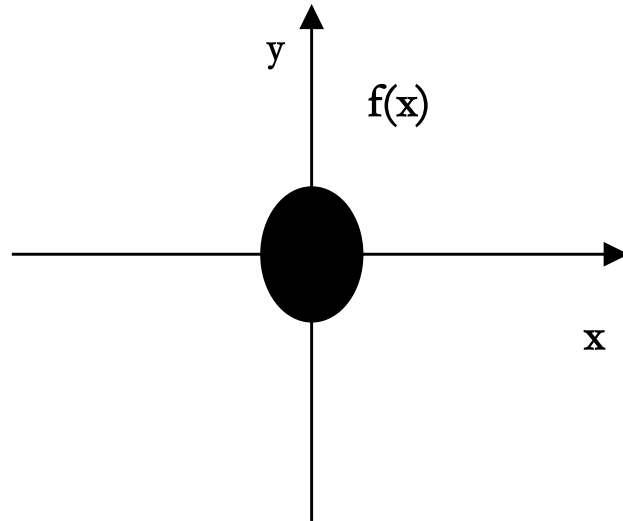
爱里斑直径 = $2.6 \mu\text{m}$

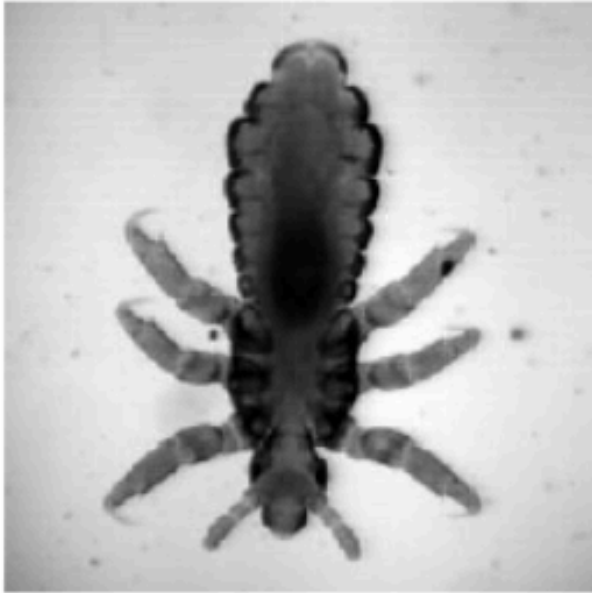
实际上一般用 $10 \sim 20 \mu\text{m}$ 的小孔即可



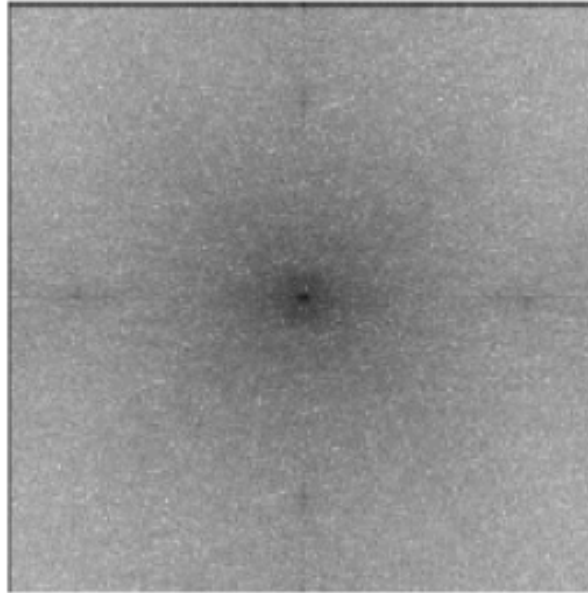
2、高通滤波：

阻挡光挡片，可突出轮廓，可处理反差低，边缘不清晰的图像。

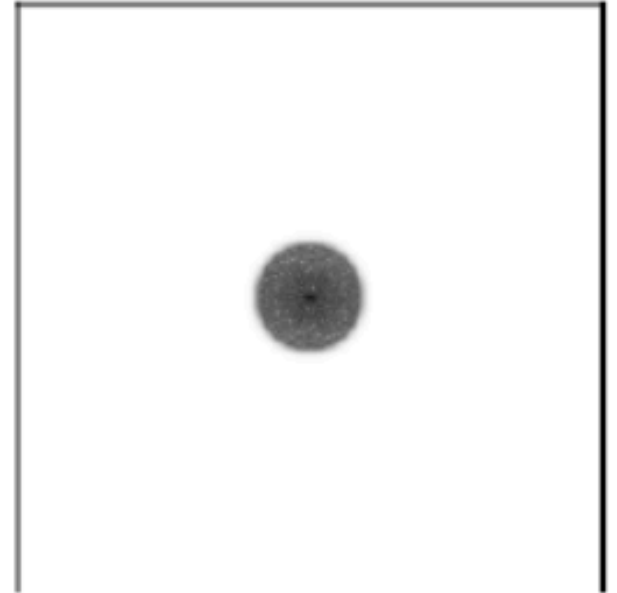




(a)



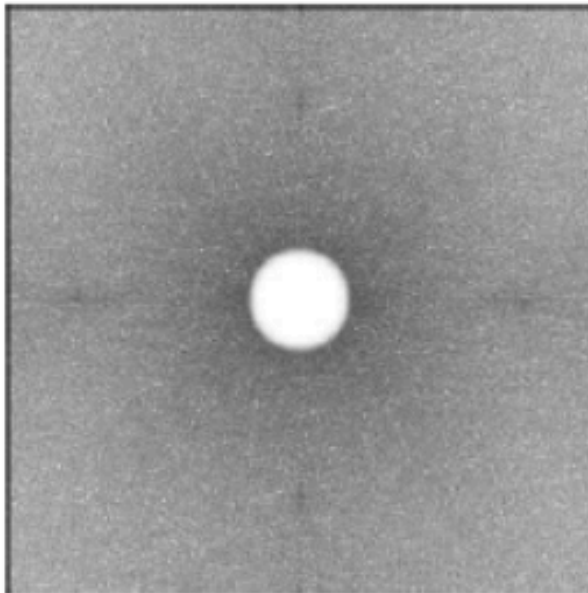
(b)



(c)



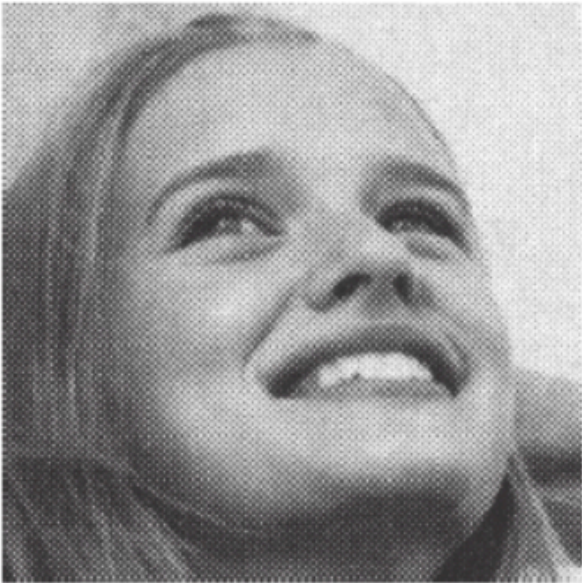
(d)



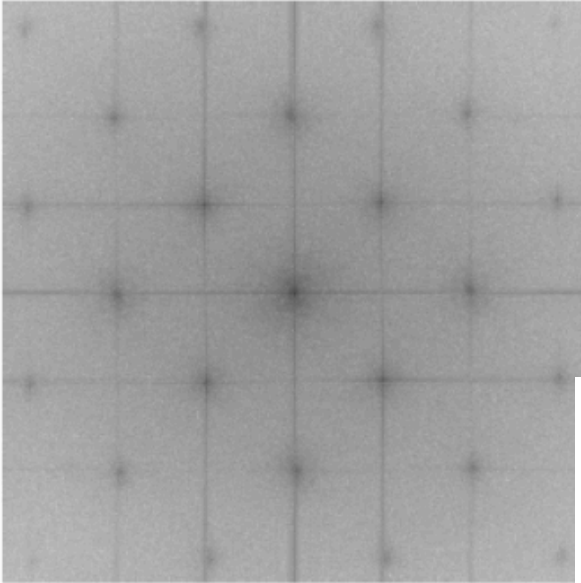
(e)



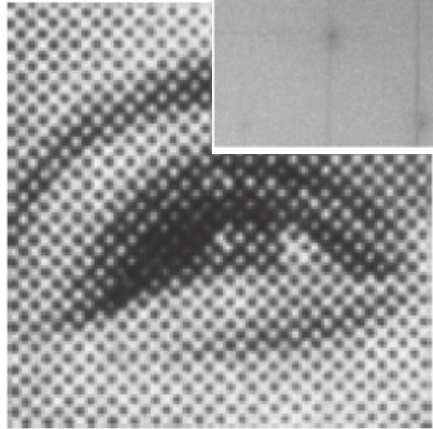
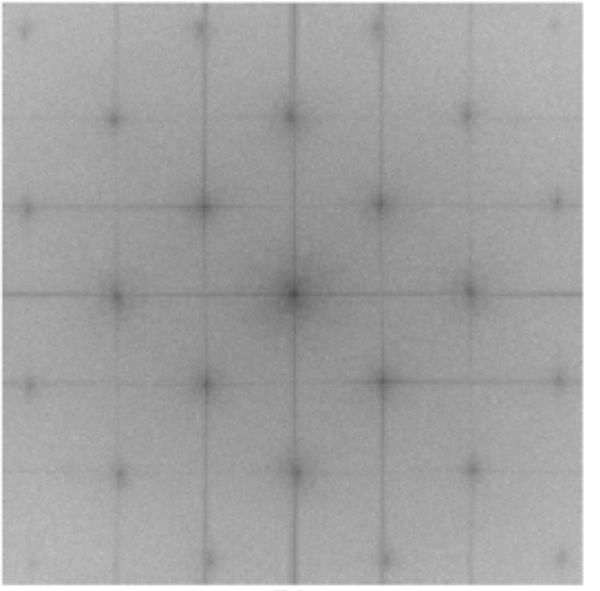
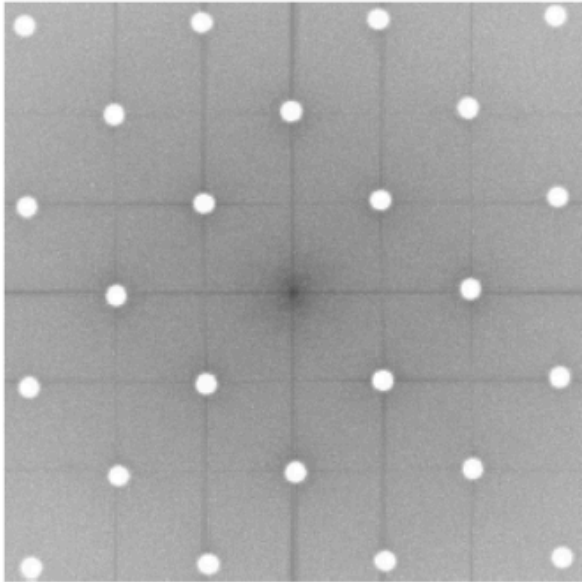
(f)



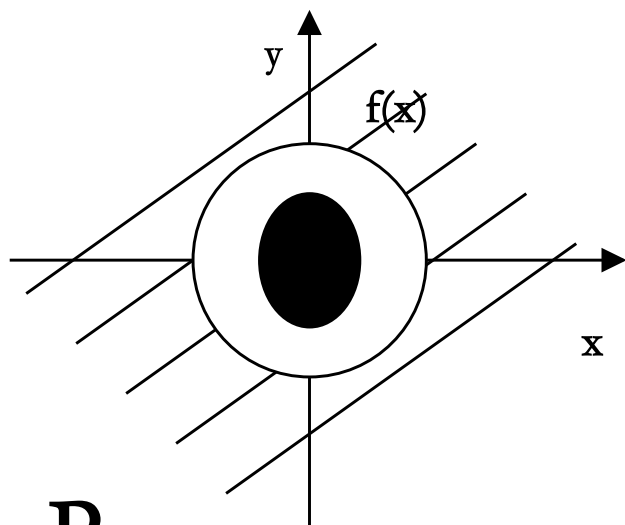
(a)



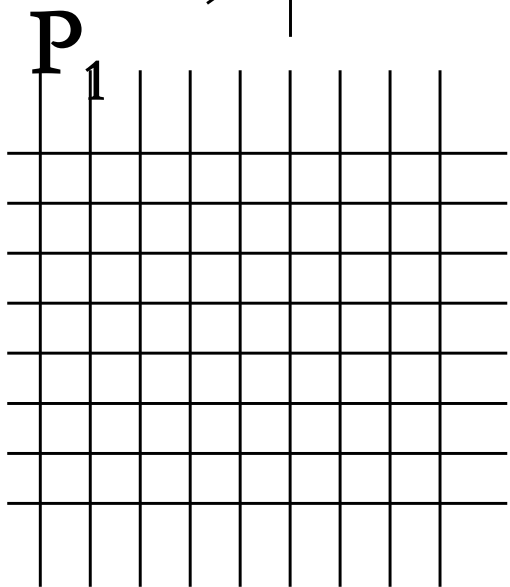
(b)



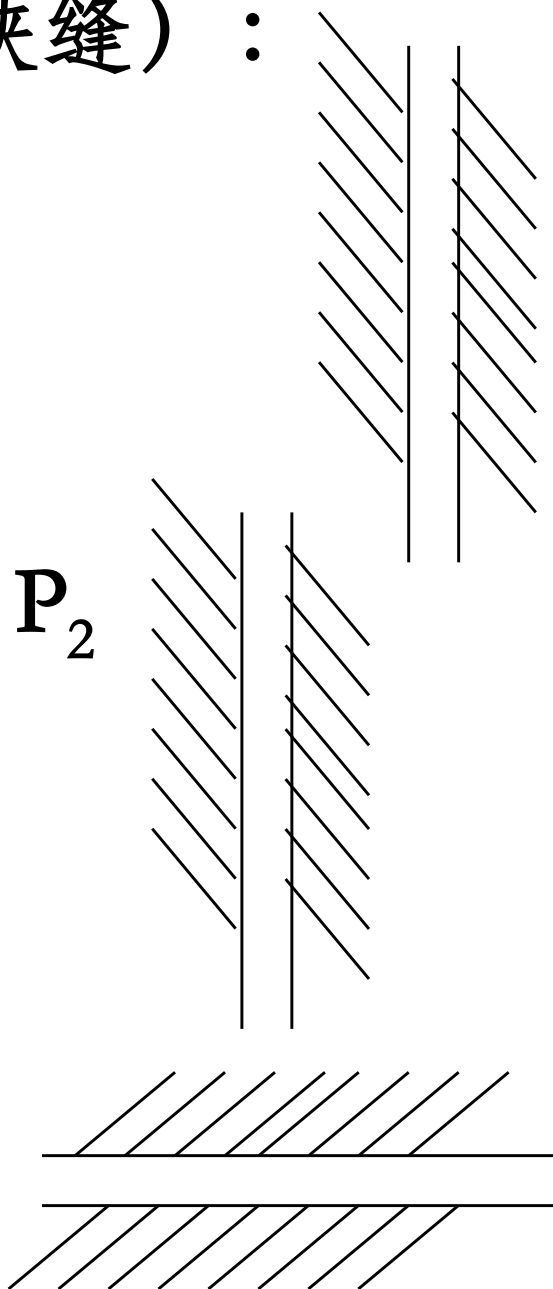
3、带通滤波（圆环或狭缝）：



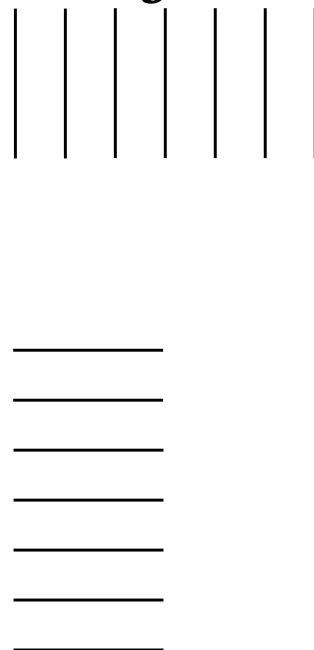
举例：

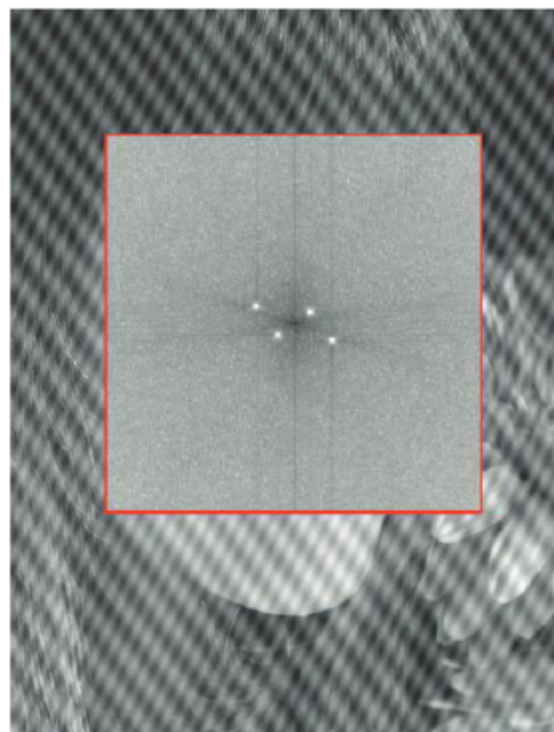
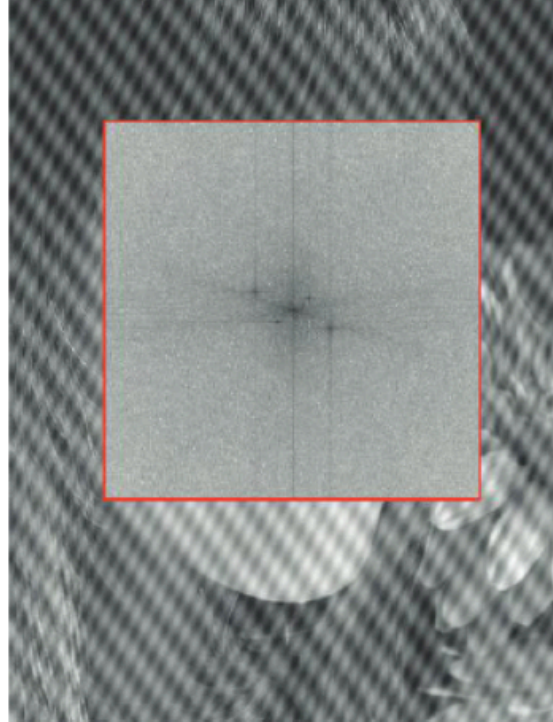


P_2



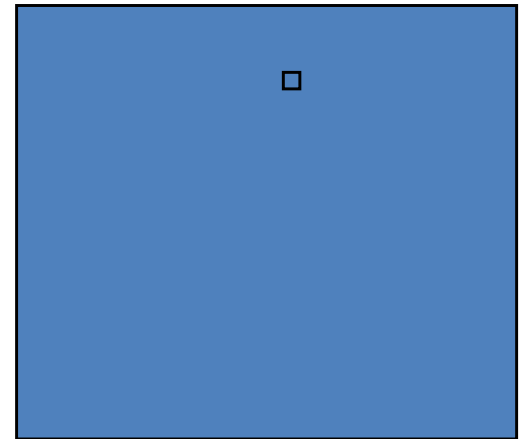
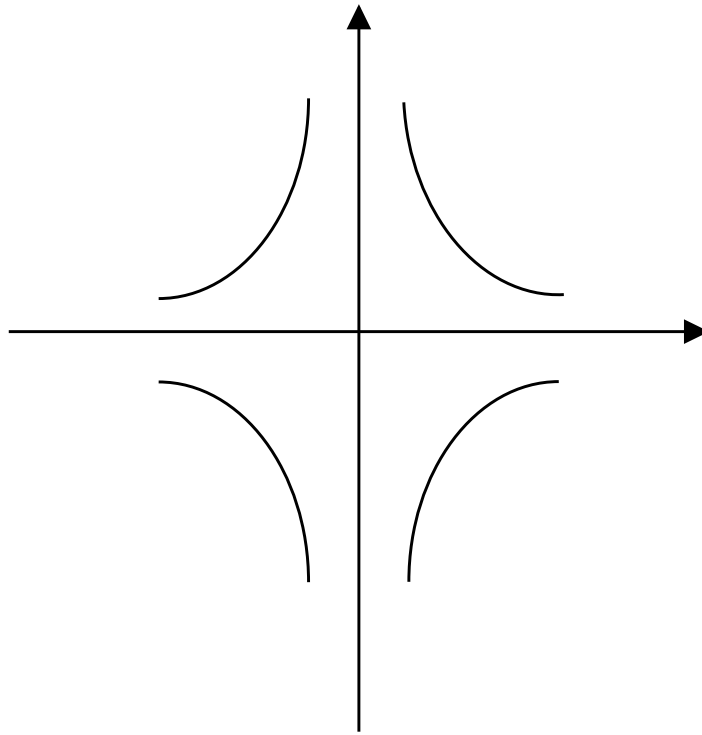
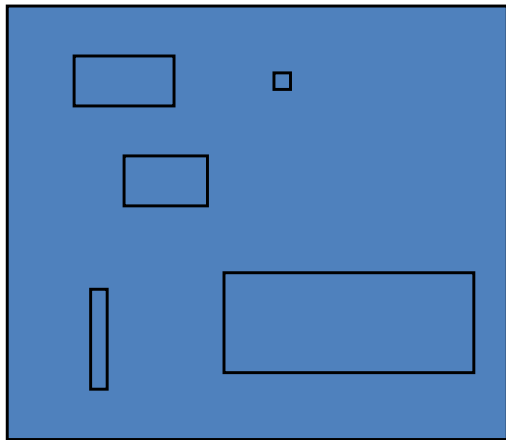
P_3



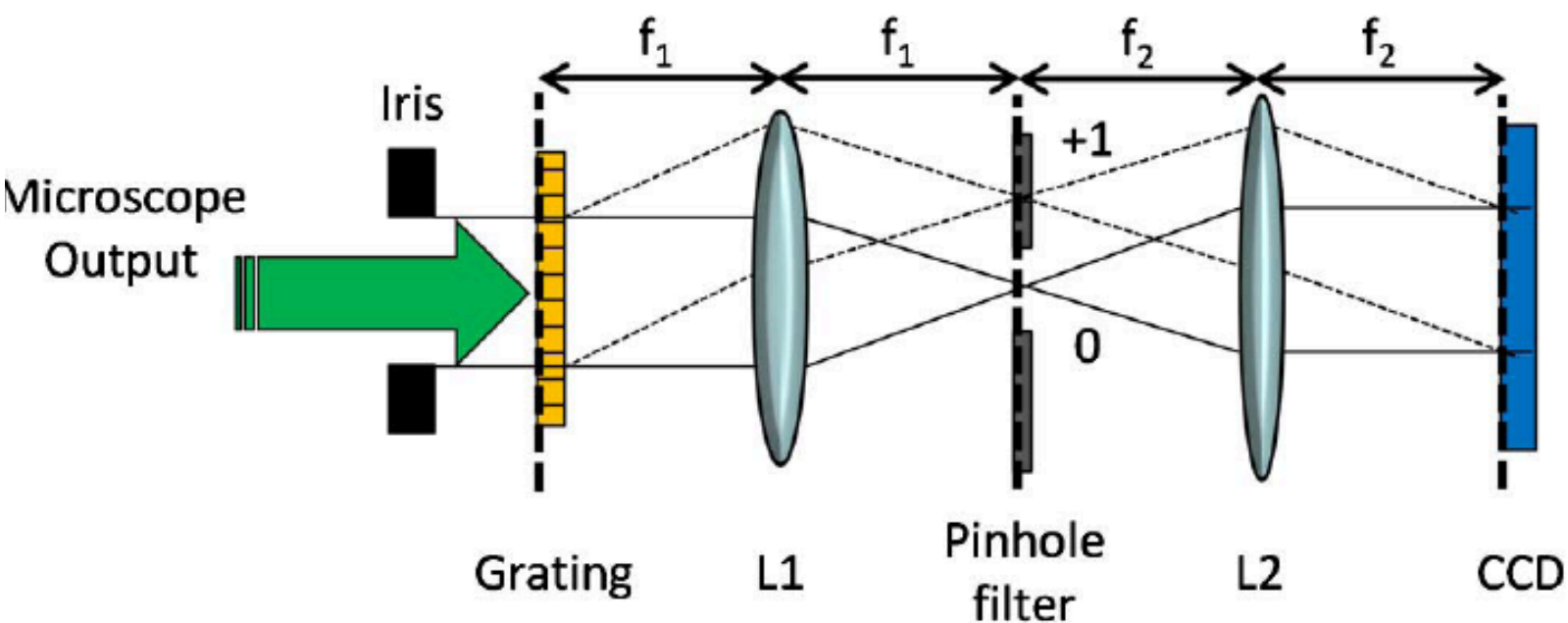


4、坏点滤波：

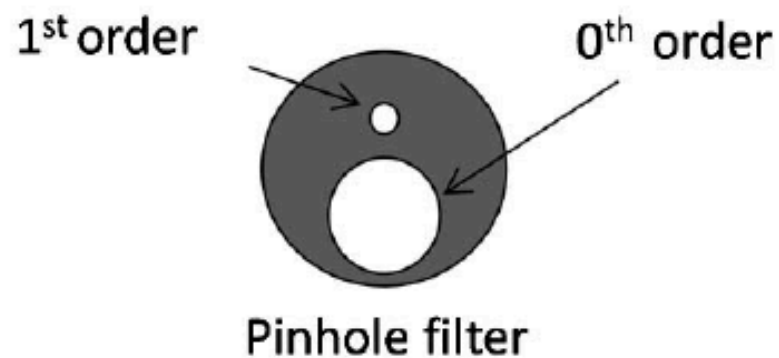
举例：集成电路掩模检查：

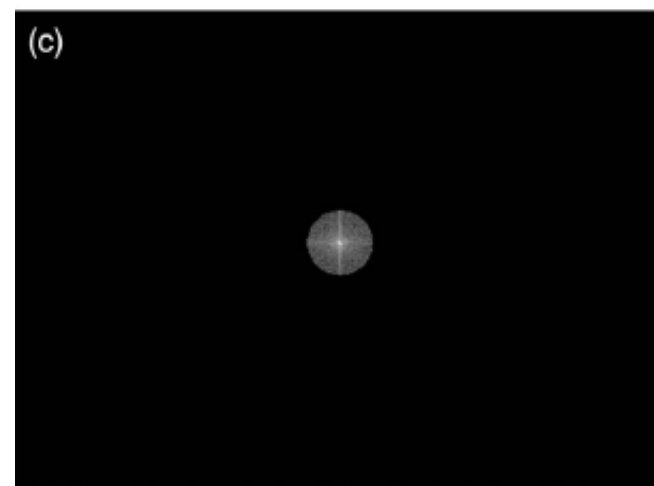
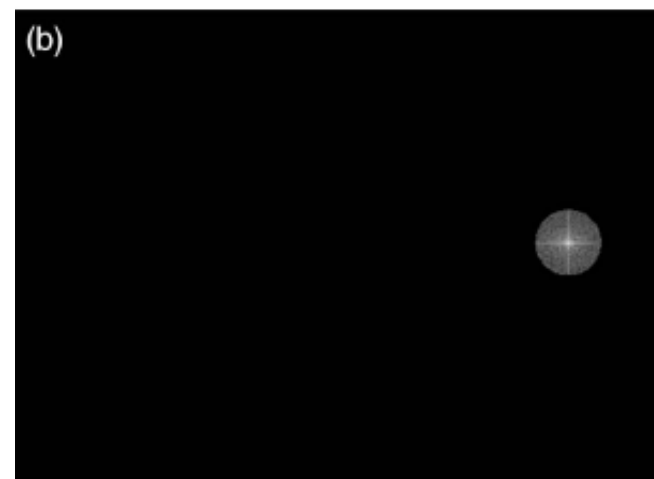
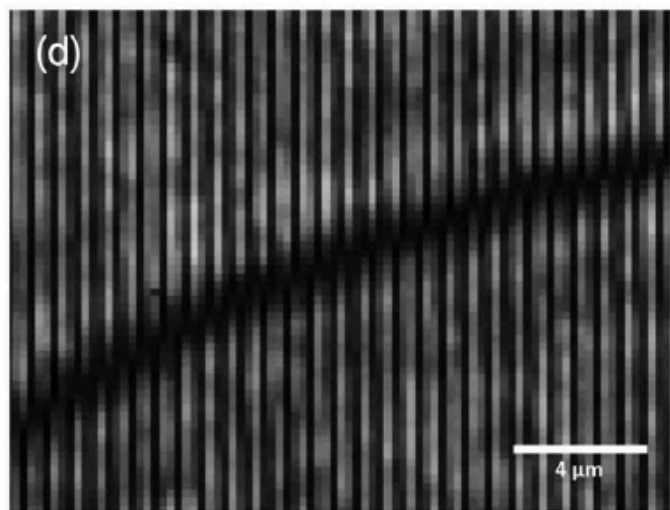
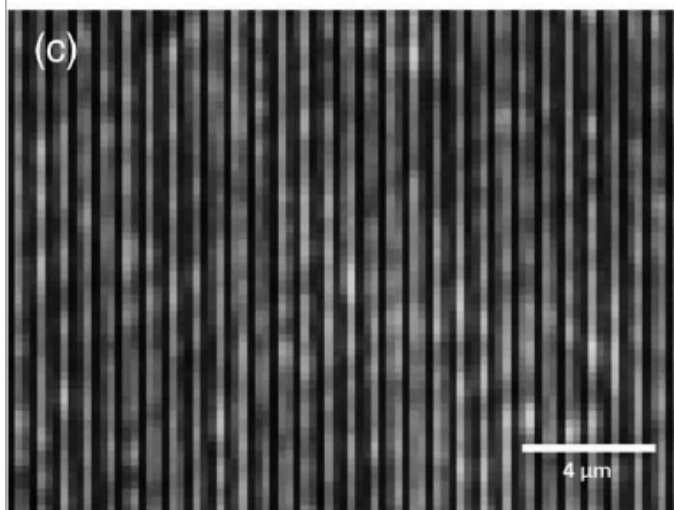
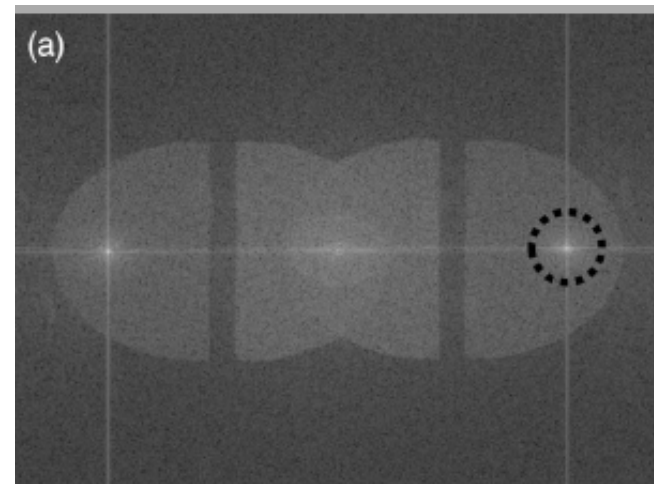
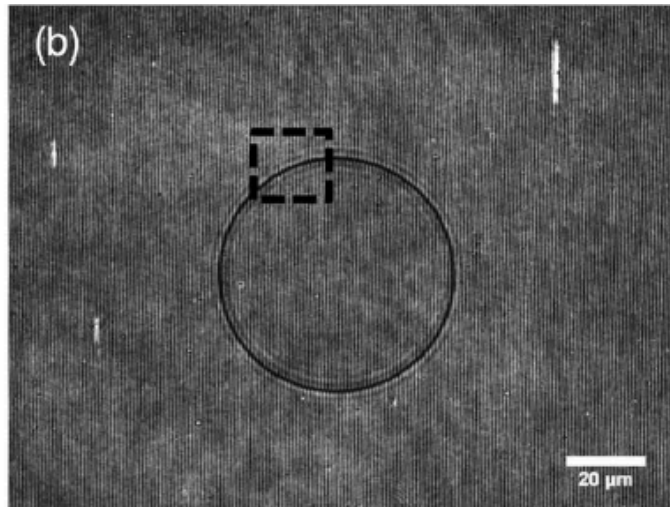
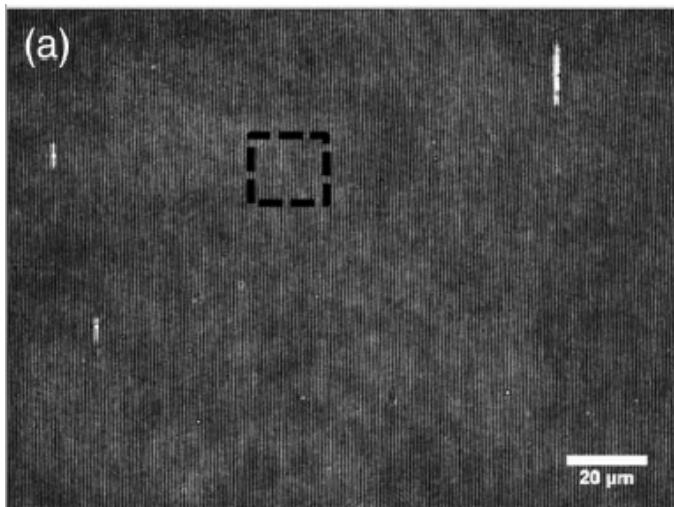


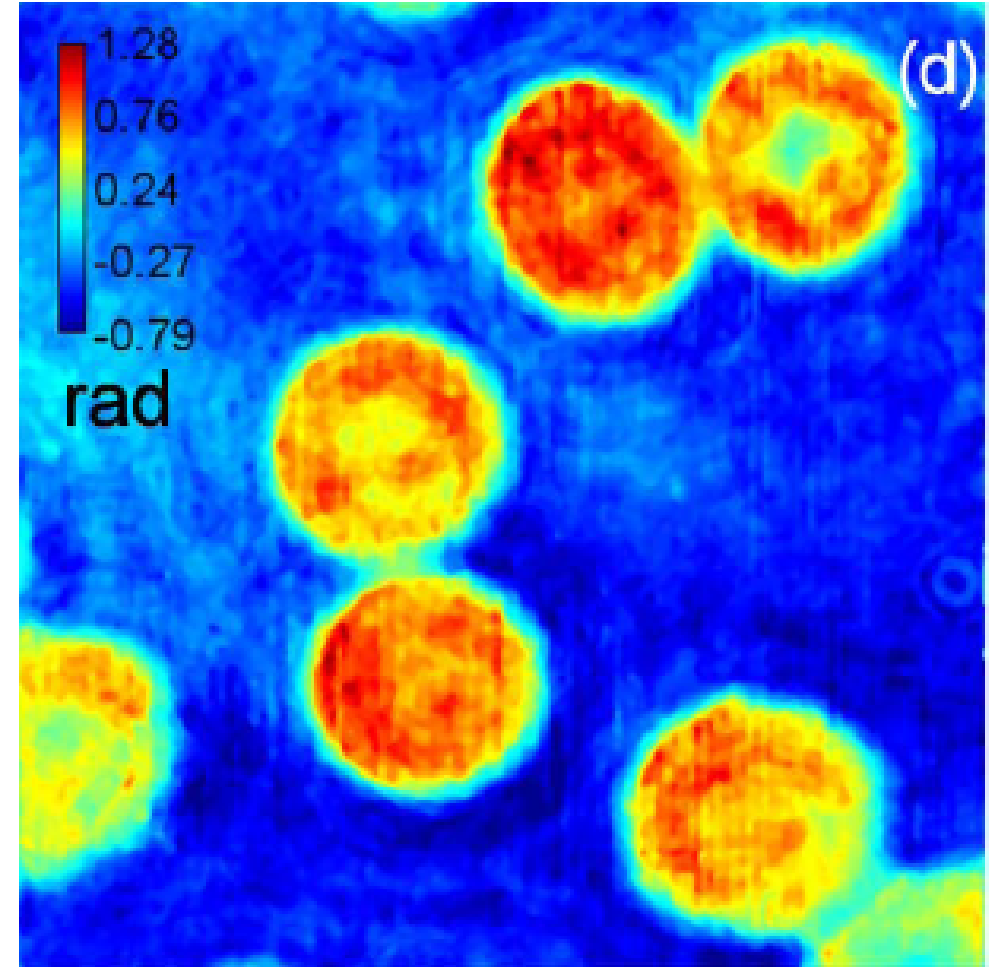
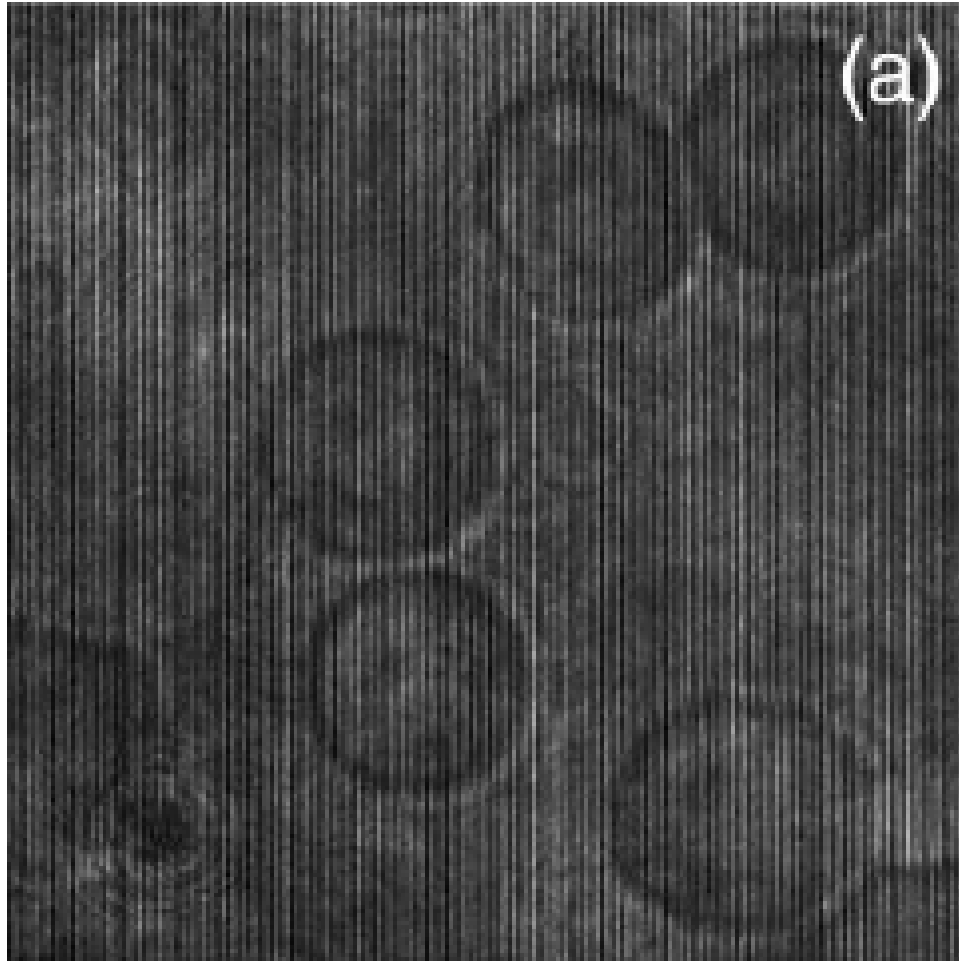
(a) Microscope Output Image Plane Fourier Plane Final Image Plane

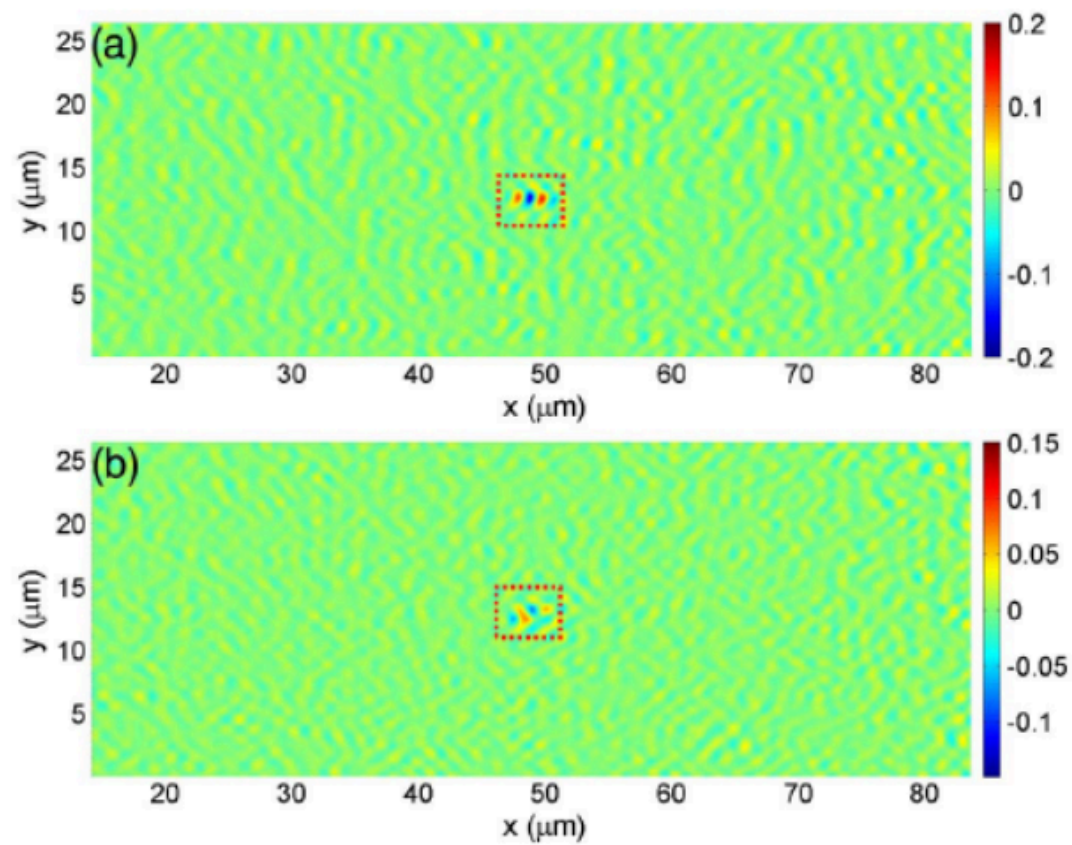
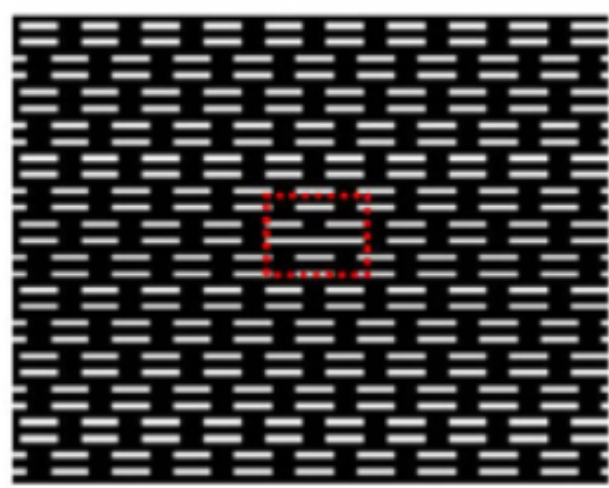
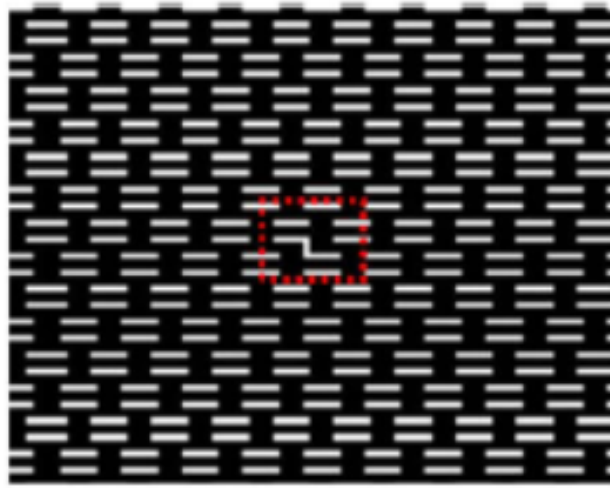
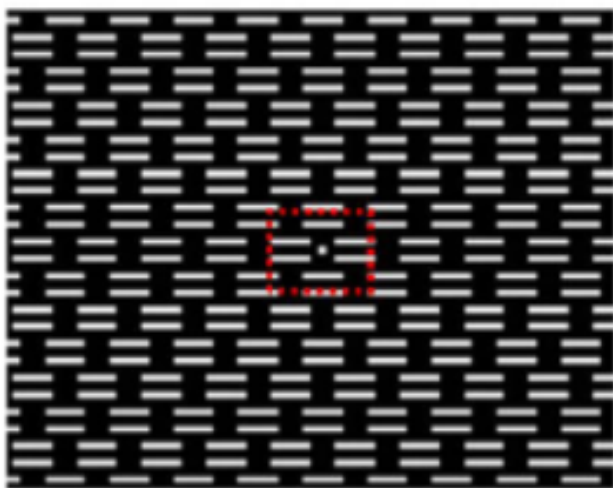


(b)





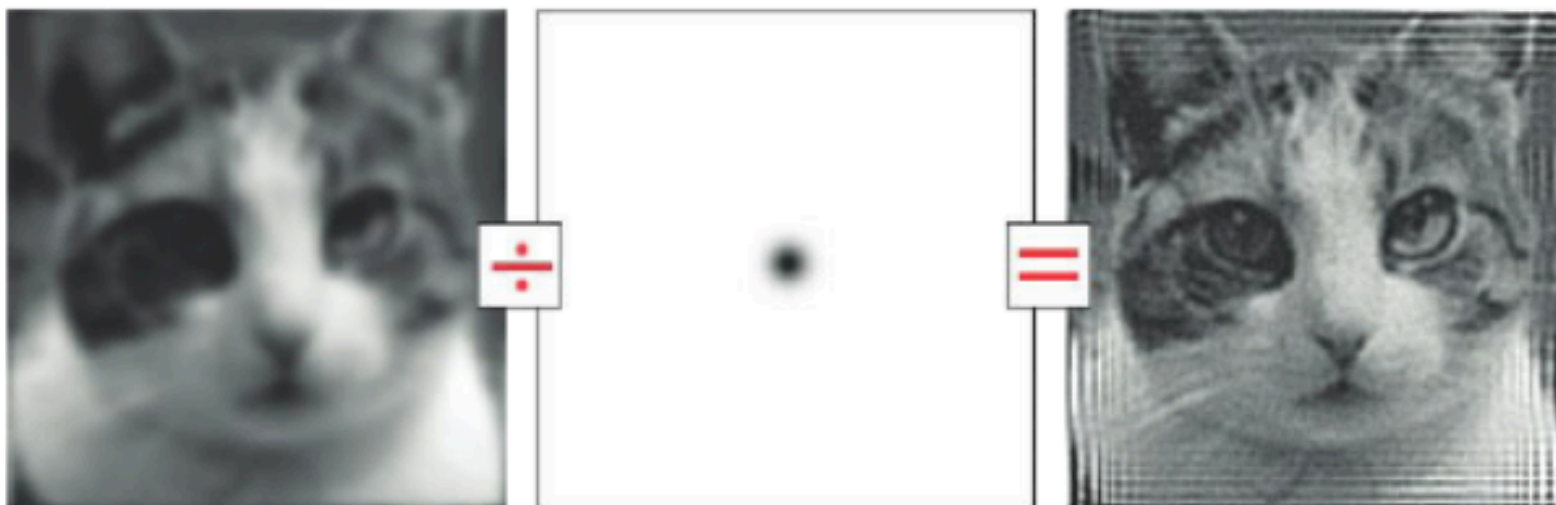




3.5.3. 全光卷积图像处理

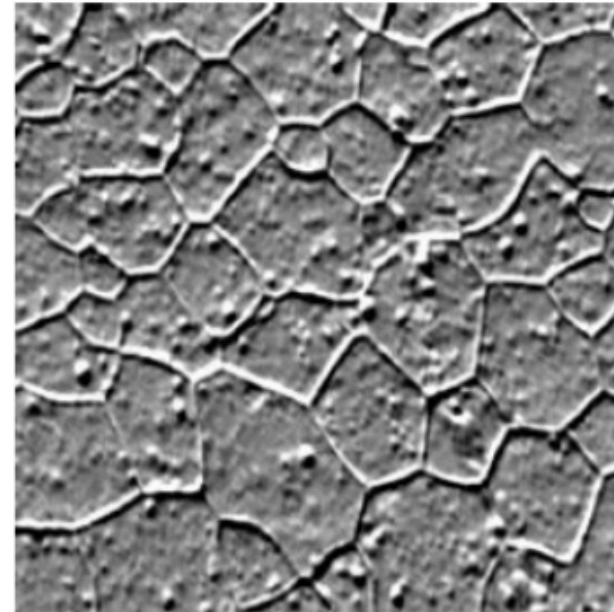
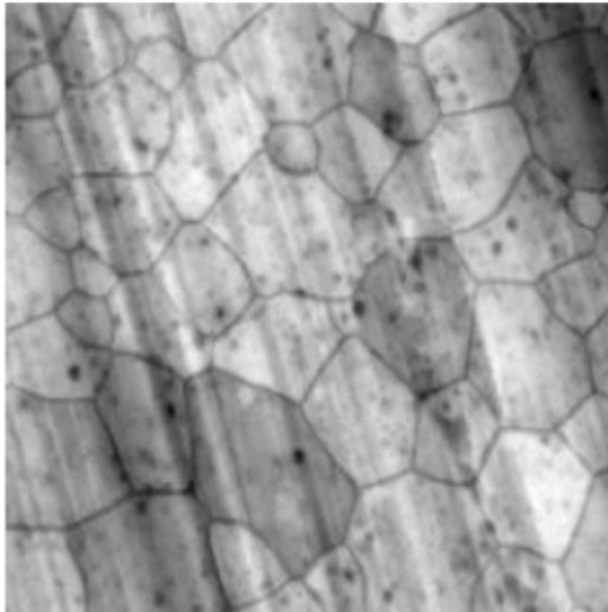


(a)



图像差分

$$h(x,y) * \text{Object}(x,y) = \text{Image}$$



$$\begin{bmatrix} +1 & 0 & -1 \\ +1 & 0 & -1 \\ +1 & 0 & -1 \end{bmatrix}$$

$$\begin{bmatrix} +1 & 0 & -1 \\ +2 & 0 & -2 \\ +1 & 0 & -1 \end{bmatrix}$$

$$\begin{bmatrix} +1 & -1 & -1 \\ +2 & +1 & -1 \\ +1 & -1 & -1 \end{bmatrix}$$

$$\begin{bmatrix} +5 & -3 & -3 \\ +5 & 0 & -3 \\ +5 & -3 & -3 \end{bmatrix}$$

$$\begin{bmatrix} +1 & +1 & 0 \\ +1 & 0 & -1 \\ 0 & -1 & -1 \end{bmatrix}$$

$$\begin{bmatrix} +2 & +1 & 0 \\ +1 & 0 & -1 \\ -1 & -1 & -2 \end{bmatrix}$$

$$\begin{bmatrix} +2 & +1 & -1 \\ +1 & +1 & -1 \\ -1 & -1 & -1 \end{bmatrix}$$

$$\begin{bmatrix} +5 & +5 & -3 \\ +5 & 0 & -3 \\ -3 & -3 & -3 \end{bmatrix}$$

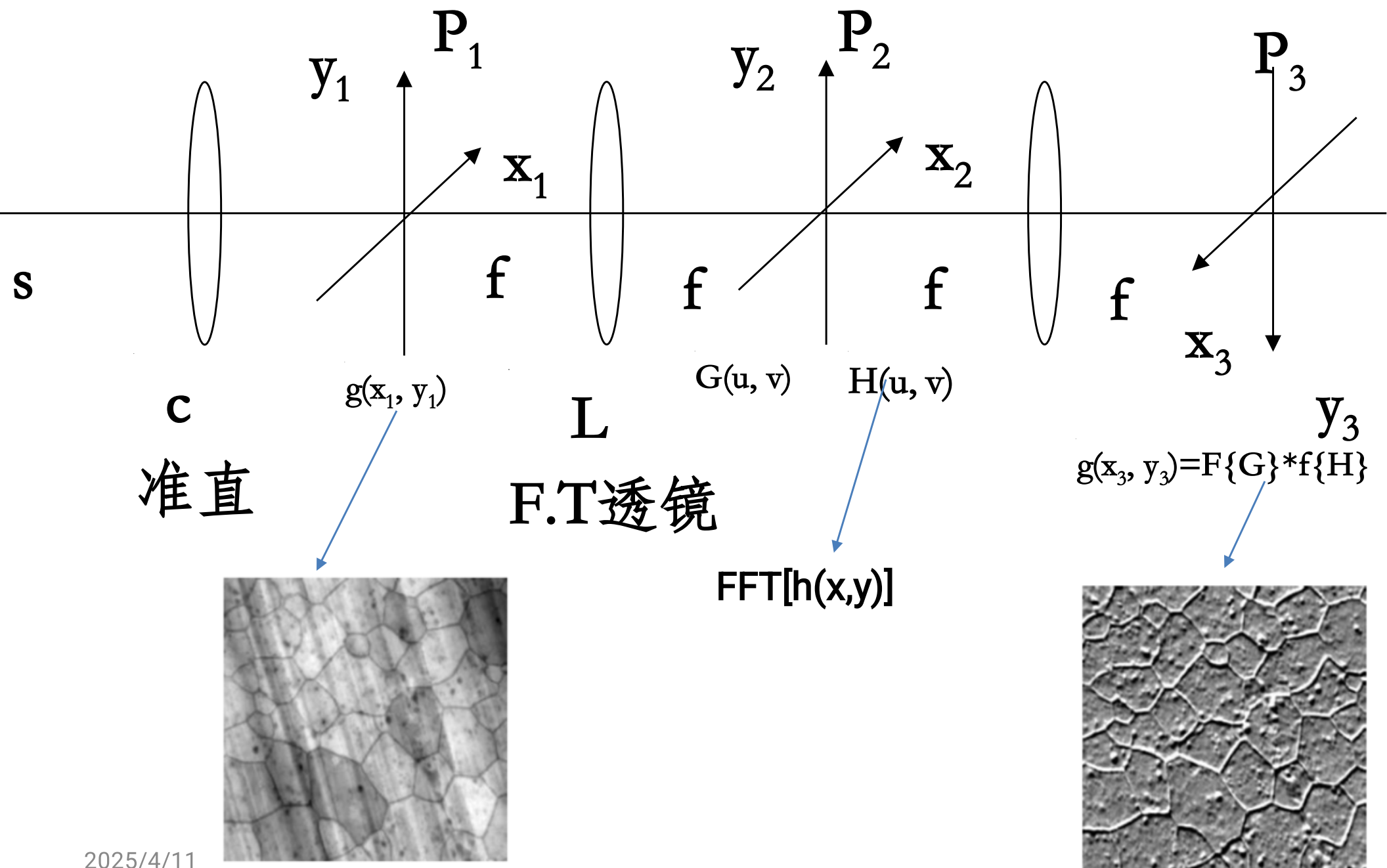
$$\begin{bmatrix} +1 & +1 & +1 \\ 0 & 0 & 0 \\ 1 & -1 & -1 \end{bmatrix}$$

$$\begin{bmatrix} +1 & +2 & +1 \\ 0 & 0 & 0 \\ -1 & -2 & -1 \end{bmatrix}$$

$$\begin{bmatrix} +1 & +2 & +1 \\ -1 & +1 & -1 \\ -1 & -1 & -1 \end{bmatrix}$$

$$\begin{bmatrix} +5 & +5 & +5 \\ -3 & 0 & -3 \\ -3 & -3 & -3 \end{bmatrix}$$

卷积图像处理的思路：



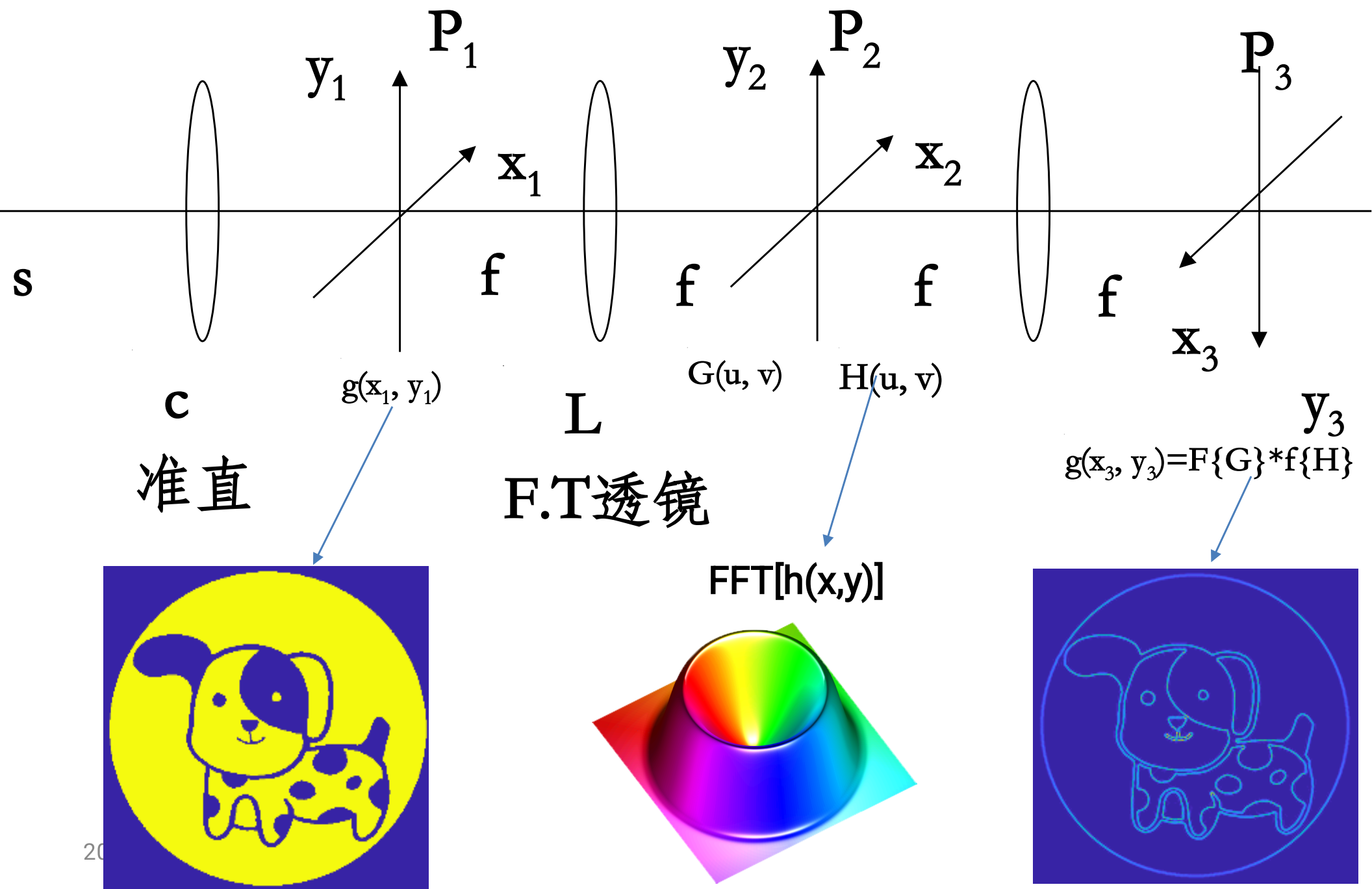
图像边缘提取



$$\otimes \begin{bmatrix} 0 & -1 & 0 \\ -1 & 4 & -1 \\ 0 & -1 & 0 \end{bmatrix} =$$



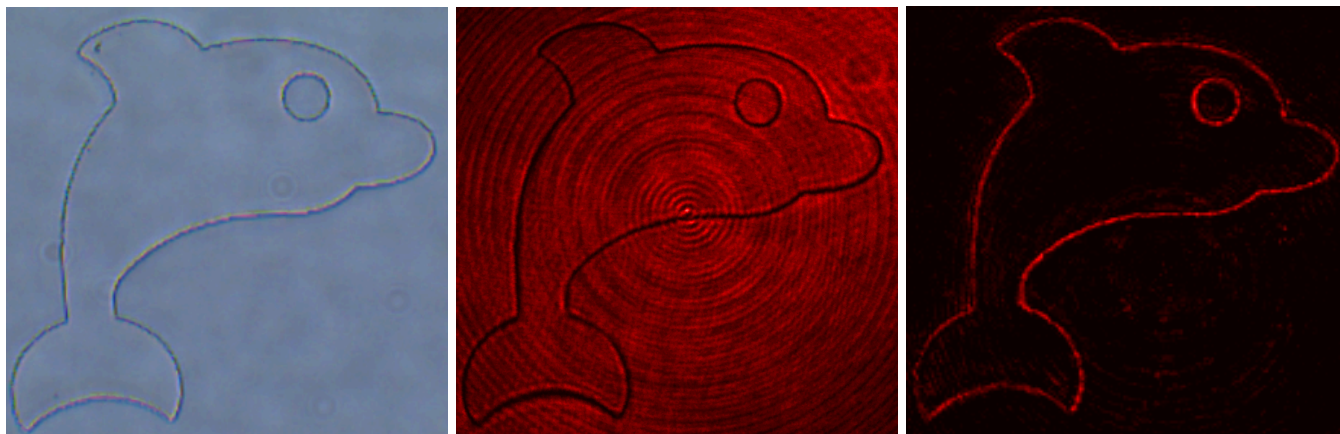
卷积图像处理的思路：



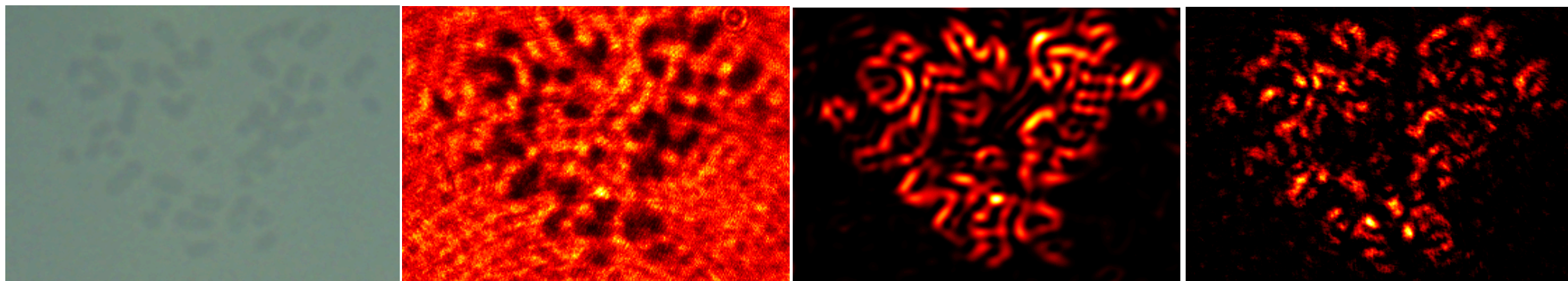
振幅型物体



相位型物体



生物样品



空间滤波器的制作方法

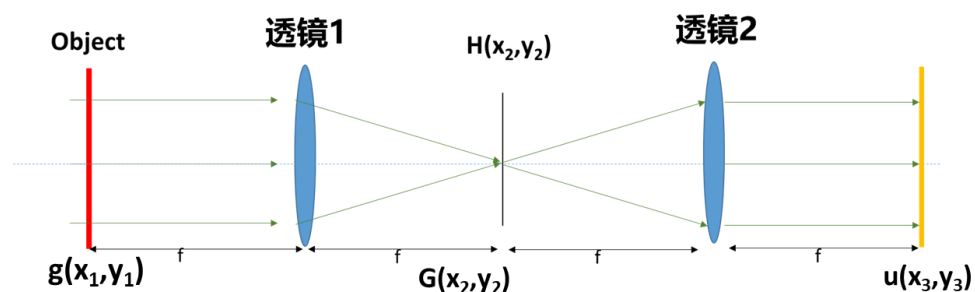
- 滤波器的制作方法：

机械法、照相法、镀膜法、光刻法...

- 1、振幅滤波器 $\varphi(u, v) = 0$, 按 $A(u, v)$ 制作
- 2、位相滤波器 $A(u, v)$ 常数, 改变 $\varphi(u, v)$
- 3、全息滤波器
- 4、计算滤波器

光学空间滤波：相位滤波器

能否处理相位物体（生物切片、油膜、温度场）？



$$|g(x_1, y_1)|^2$$

物体：

$$g(x_1, y_1) = e^{i\varphi(x_1, y_1)} \approx \overrightarrow{1 + i\varphi(x_1, y_1)}$$

频谱面：

滤波器：

$$\boxed{\bullet} \rightarrow \mathbf{G}(x_2, y_2) = \delta(x_2, y_2) + \overrightarrow{i\Phi(x_2, y_2)}$$

$$\mathbf{G}(x_2, y_2) * \mathbf{H}(x_2, y_2) = i\Phi(x_2, y_2)$$

光学空间滤波：相位滤波器

物体：

$$g(x_1, y_1) = e^{i\varphi(x_1, y_1)} \approx 1 + i\varphi(x_1, y_1)$$

频谱面：

$$G(x_2, y_2) = \delta(x_2, y_2) + i\Phi(x_2, y_2)$$

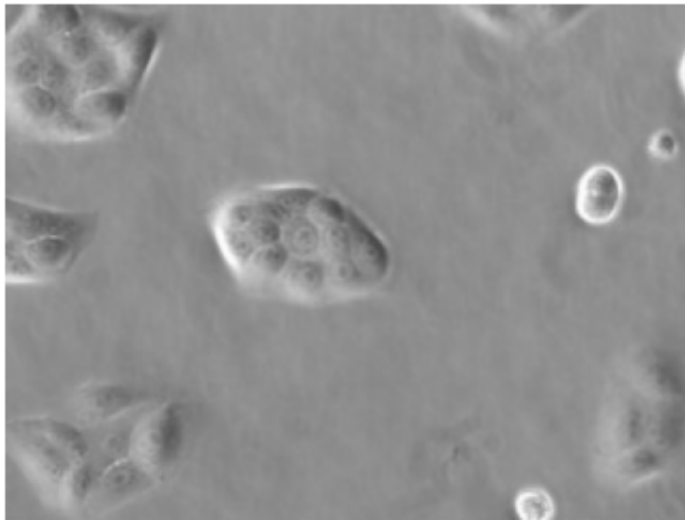
相位滤波器：  →

$$G(x_2, y_2) * H(x_2, y_2) = i\delta(x_2, y_2) + i\Phi(x_2, y_2)$$



$$U(x_3, y_3) = i + i\varphi(-x_3, -y_3)$$

$$I(x_3, y_3) = 1 + 2\varphi(-x_3, -y_3) + \dots$$



光学空间滤波：相位滤波器

泽尼克（1888-1966），荷兰物理学家，因论证相衬法，特别是发明了相衬显微镜而获诺贝尔奖（1953年）。他制造了世界上第一台相衬显微镜。



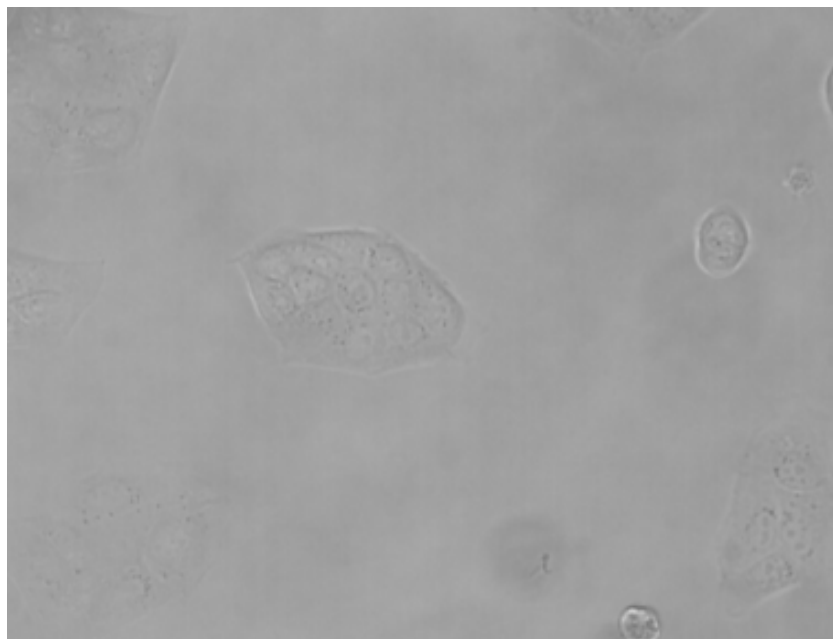
Frederik Zernike



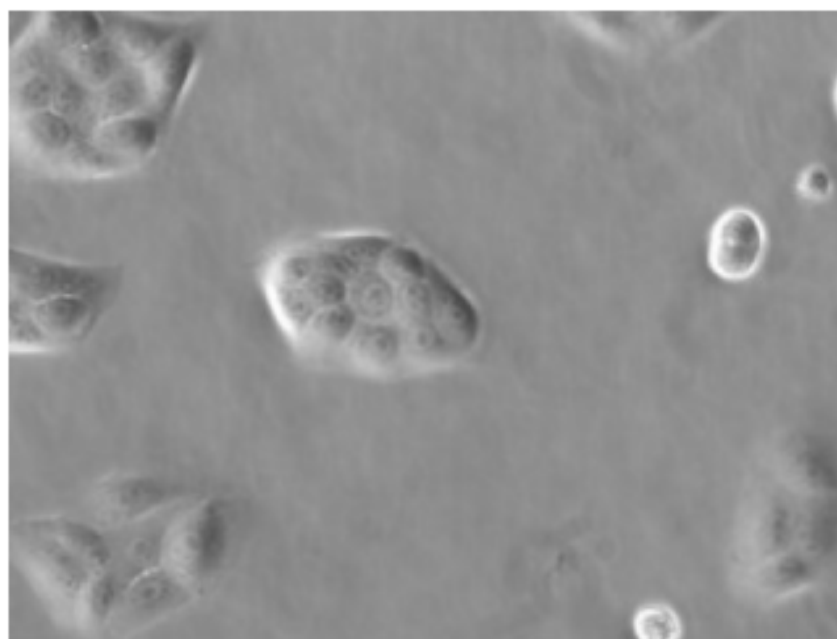
相衬显微镜

相衬法是通过空间滤波器将物体的位相信息转换为相应的振幅信息，从而大大提高透明物体的可分辨性，所以从这个意义上说，相衬法是**最早的光学信息处理**方法之一，在光学的发展史上具有重要意义。

普通显微镜



相衬显微镜



扩展延伸

光学空间滤波：相位滤波器

$$I(x_3, y_3) = 1 + \varphi(-x_3, -y_3) + \varphi^2(-x_3, -y_3)$$

如何提升图像对比度？

频谱面：
$$G(x_2, y_2) = \delta(x_2, y_2) + i\Phi(x_2, y_2)$$

$$G(x_2, y_2) * H(x_2, y_2) = i \cdot t \cdot \delta(x_2, y_2) + i\Phi(x_2, y_2)$$

成像面：

$$U(x_3, y_3) = it + i\varphi(-x_3, -y_3)$$

$$I(x_3, y_3) = t^2 + 2t\varphi(-x_3, -y_3) + \varphi^2(-x_3, -y_3)$$

8
0

对比度： $\mu = 2\varphi/t$ 其中 $t \ll 1$